

How to feed Machine Learning algorithms with proper metrics?

SC2.28

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e-learning
SOGA

<https://www.geo.fu-berlin.de/en/v/soga/>



github

https://github.com/soga-lab/EGU2026_SC

Outline

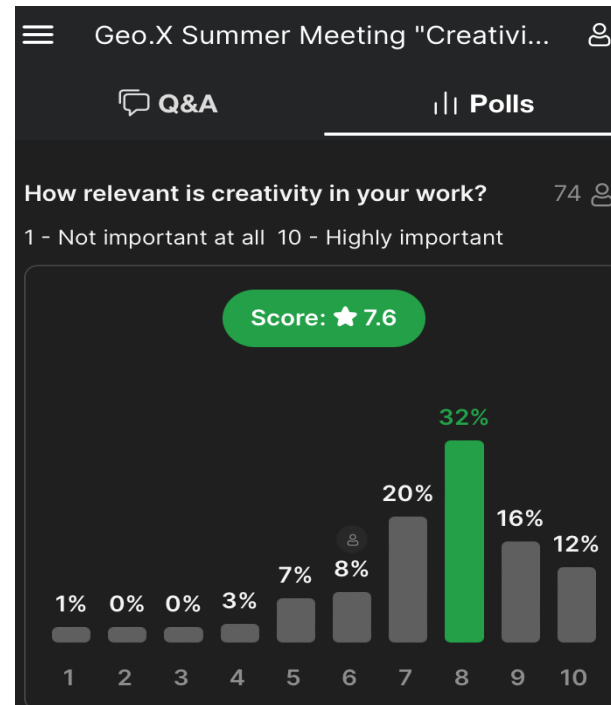
Slides with such blue background are for deeper insights or details
→ not part of the oral!

0. Motivation
1. Digesting Measures: How does MLA decide?
2. Meaningful measures: Truth beyond stats textbooks
3. Recipes: Preparing real-world data for ML
4. Hands-on exercises

Meaningless measures

3

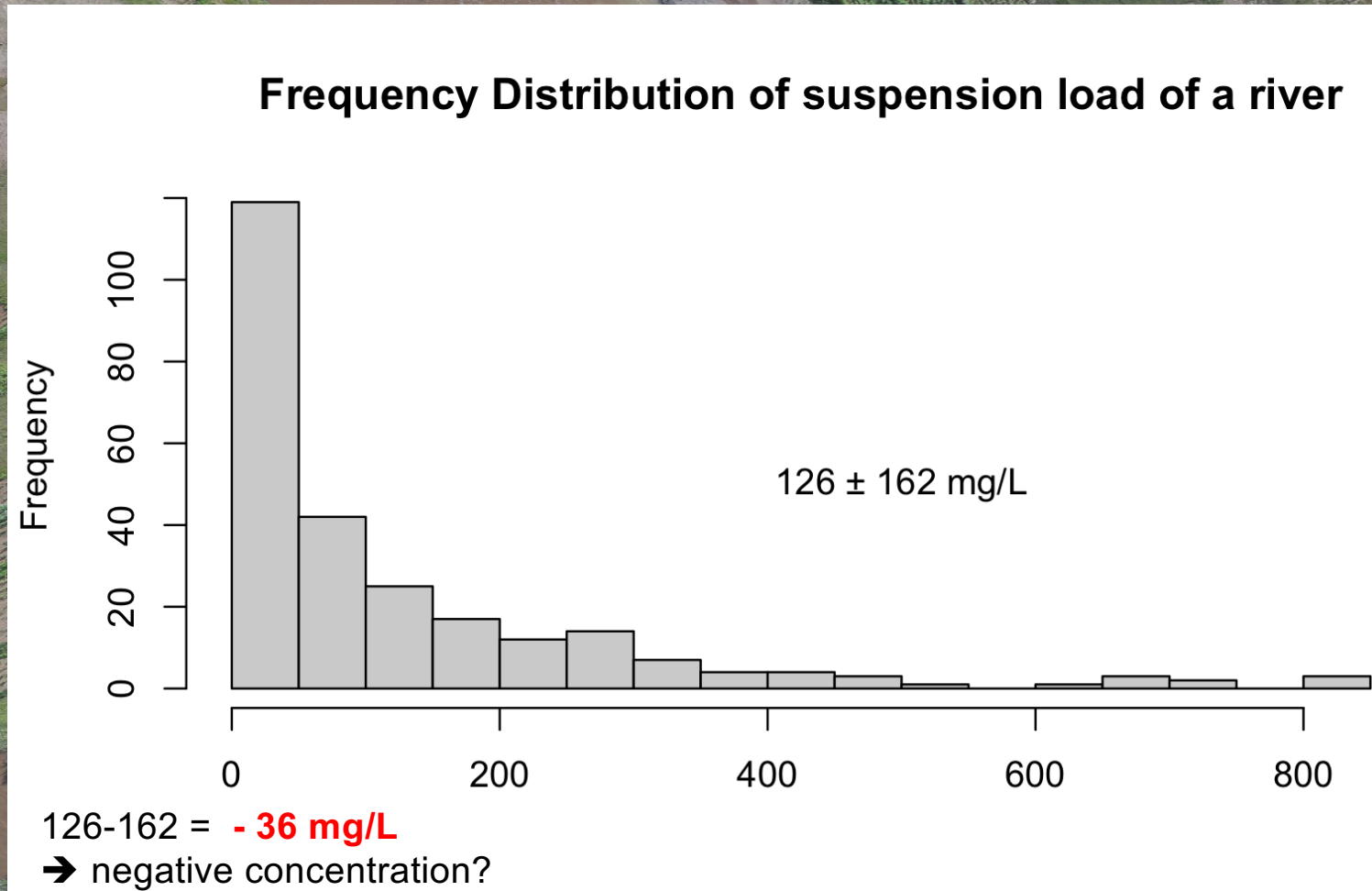
Once upon a time on a Geoscientific community meeting:



The suggested centre is part of the minority!!!!

Meaningless measures

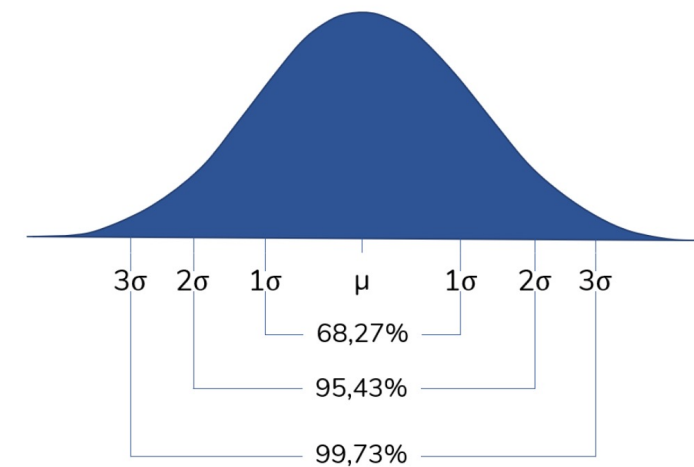
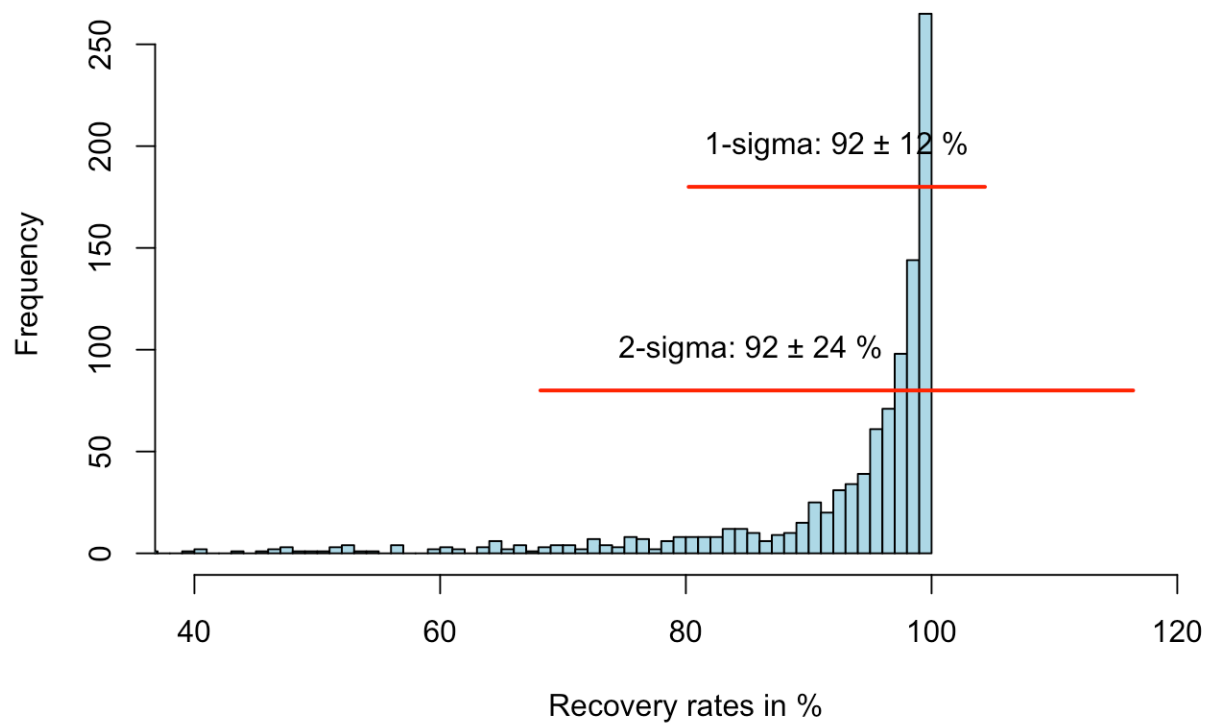
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Meaningless measures

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Percent area recovered within 30 years



- Are these very special samples with unfavorable measurements?
- Just an exception?

Meaningless measures

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$n = 19$						
	Range			Mean $\pm$ sd		
	2017	2018	2019	2017	2018	2019
	Dry	Wet	Wet	Dry	Wet	Wet
Temp	20.1–31.2	23.3–33.2	22.4–32.1	25.0 $\pm$ 3.3	28.8 $\pm$ 2.2	27.2 $\pm$ 2.8
pH	7.3–9.4	6.3–8.6	7.0–9.2	8.2 $\pm$ 0.6	7.4 $\pm$ 0.8	8.1 $\pm$ 0.7
EC	80–2230	90–2020	76–2935	911.2 $\pm$ 669.9	614.7 $\pm$ 655.4	724.9 $\pm$ 229.3
Turbidity	17.5–1471.6	47.9–1422.3	16.1–1615.4	631.5 $\pm$ 551.6	722.2 $\pm$ 545.9	477.6 $\pm$ 496.7
Cl [–]	3.9–554.0	9.28–678.0	3.9–342.0	206.9 $\pm$ 179.0	170 $\pm$ 209.1	71.4 $\pm$ 105.4
SO ₄ ^{2–}	72.5–707.1	48.7–1224.0	47.0–309.2	252 $\pm$ 171.1	298 $\pm$ 334.3	126.1 $\pm$ 78.9
NO ₃ [–]	0.1–3.5	1.6–24.1	0.2–3.8	1.6 $\pm$ 1.1	8.5 $\pm$ 6.9	1.0 $\pm$ 1.1

- “I reported my data in this way, **because everybody is doing this.**”
- “I want to keep my data in an **comparable way**”!
- “This just happened because of **the large variance!**”
- “The **sample size was too small** for a more accurate estimation!”
- “Don’t take this too serious, it is **scientific data reporting standard!**”
- “I just wanted to **show the mean and the variability** of my data!”
- “I’ve tried the statistical right way, but then **my story has disappeared!**”

Did you heard another excuse? Please share it with us in the comments!

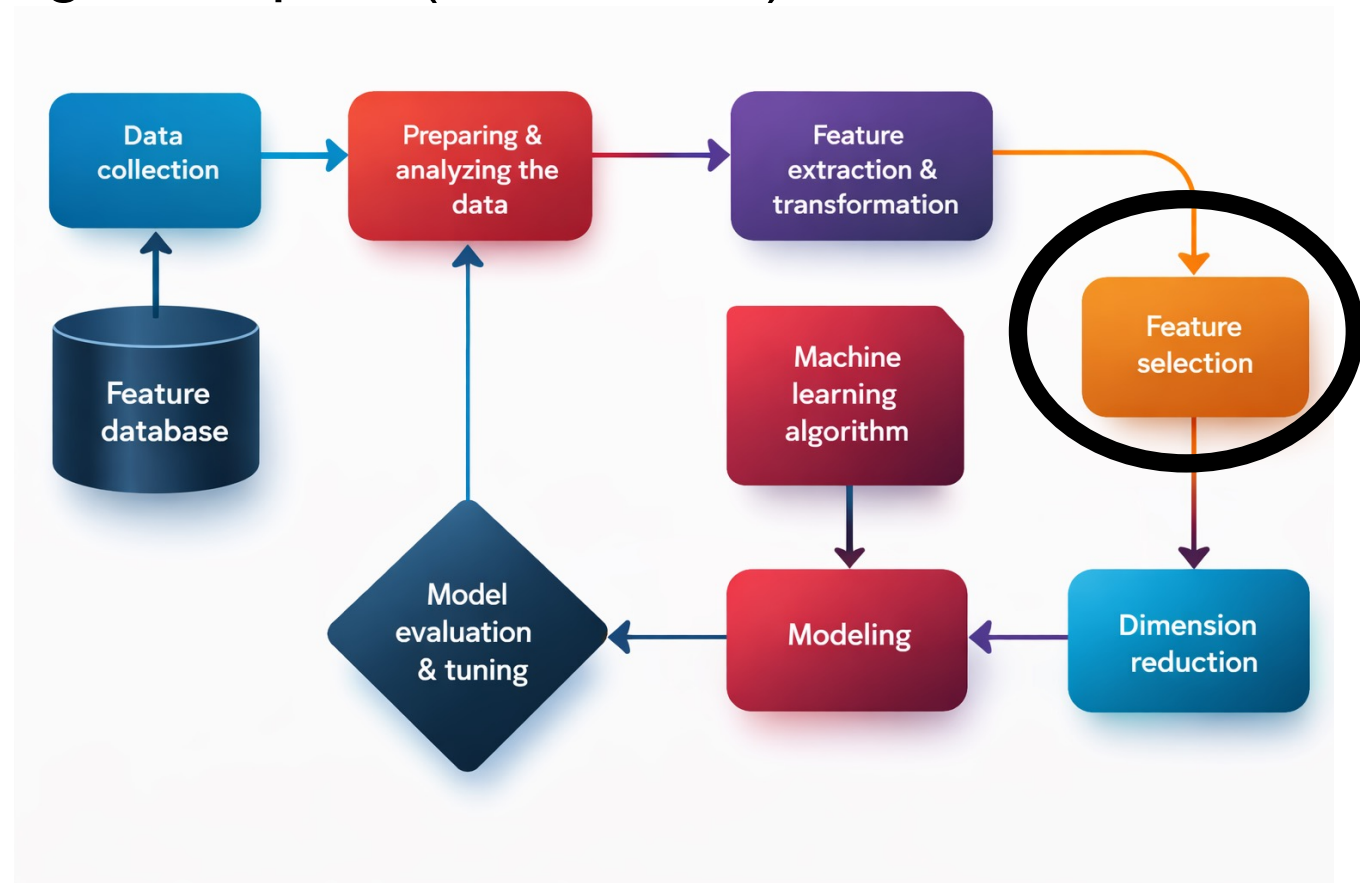
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How does MLA decide?

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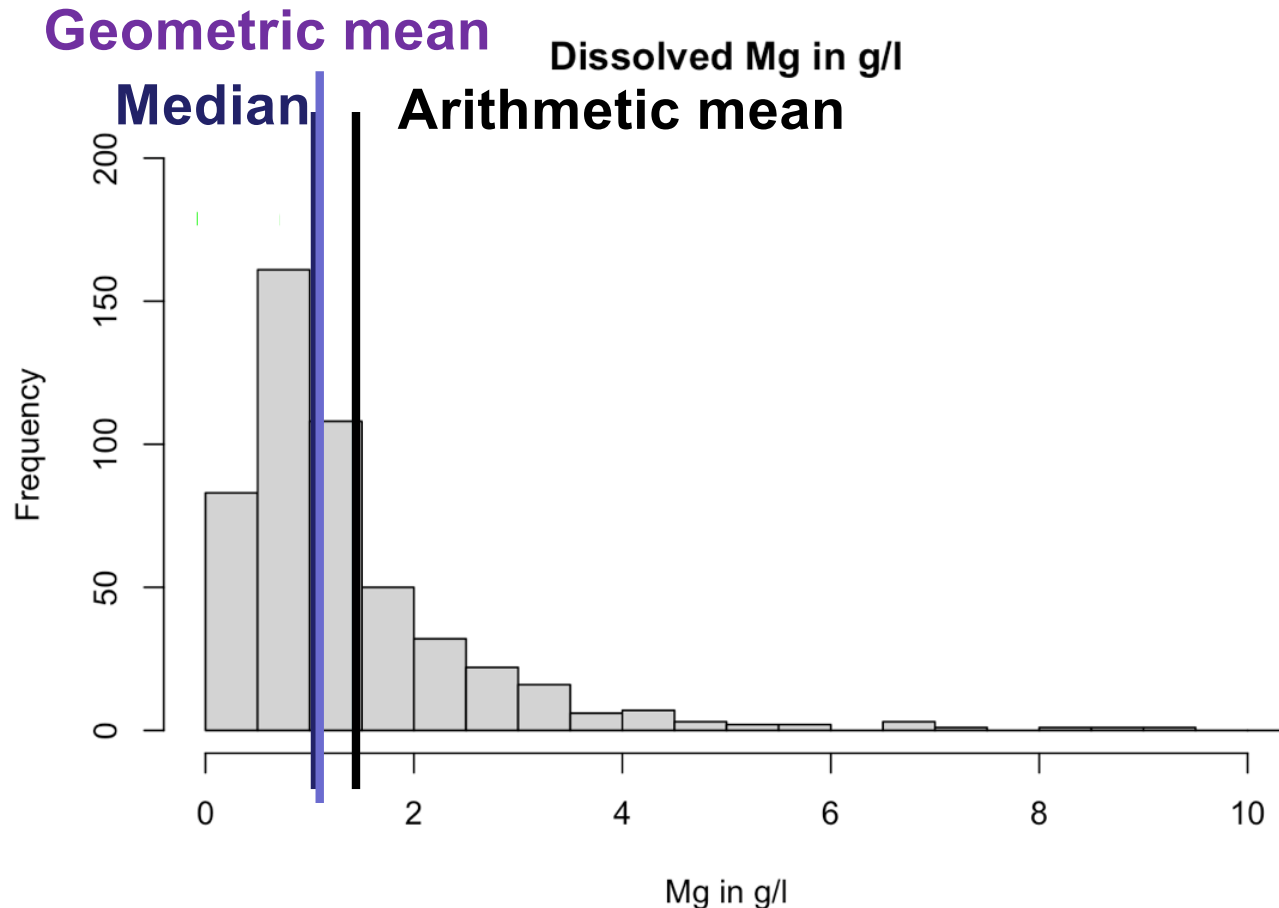
1. Choosing/Scaling the Inputs (= Features)



1. Digesting Measures: How does MLA decide?

Why do we need proper metrics?

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- **Median** **1.05**
- **Geometric mean** **1.04**
- **Arithmetic mean** **1.42**
 $\rightarrow \mu - 2\sigma < 0$

$\rightarrow$ **Why is this a problem in the application of ML algorithms?**

Why do we need proper metrics?

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- Given paired data points of n pairs $\{(x_1, y_1), \dots, (x_n, y_n)\}$
- The Pearson Correlation Coefficient is defined as:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} = \frac{s_{xy}}{s_x s_y}$$

Arithm. Mean

Covariance

Standard Deviations of x and y

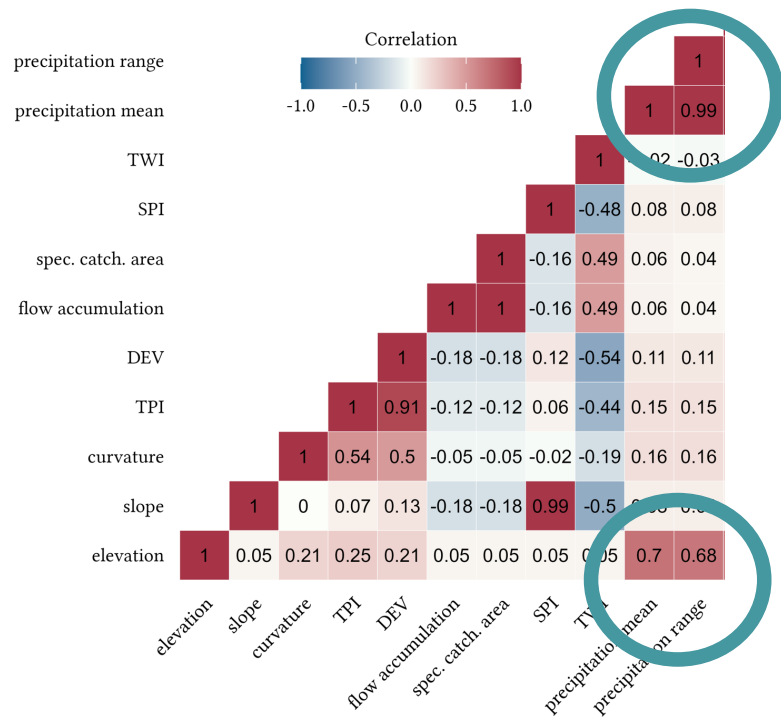
- → Spurious correlations without FST

Why do we need proper metrics?

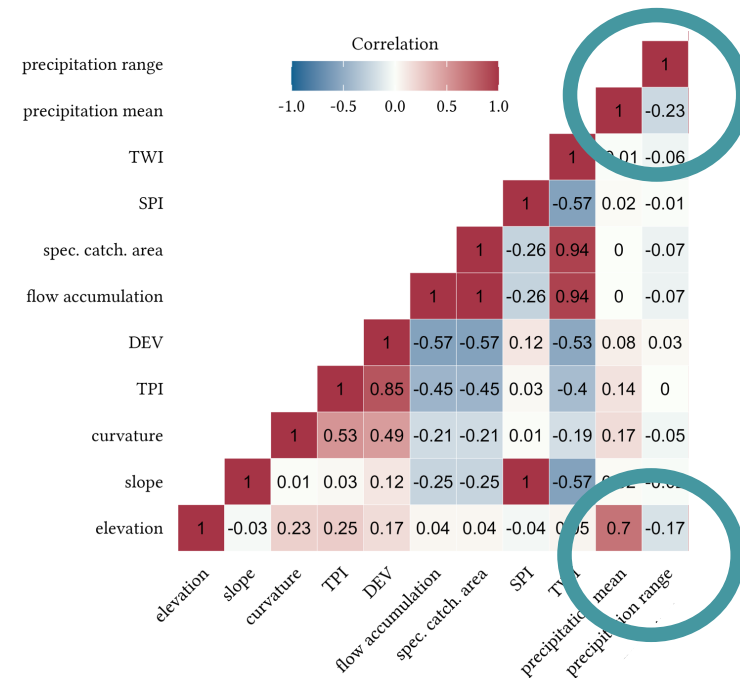
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- Example: Gully Susceptibility in Lesotho (Frenzel et al, in prep.)

Before FST



After FST

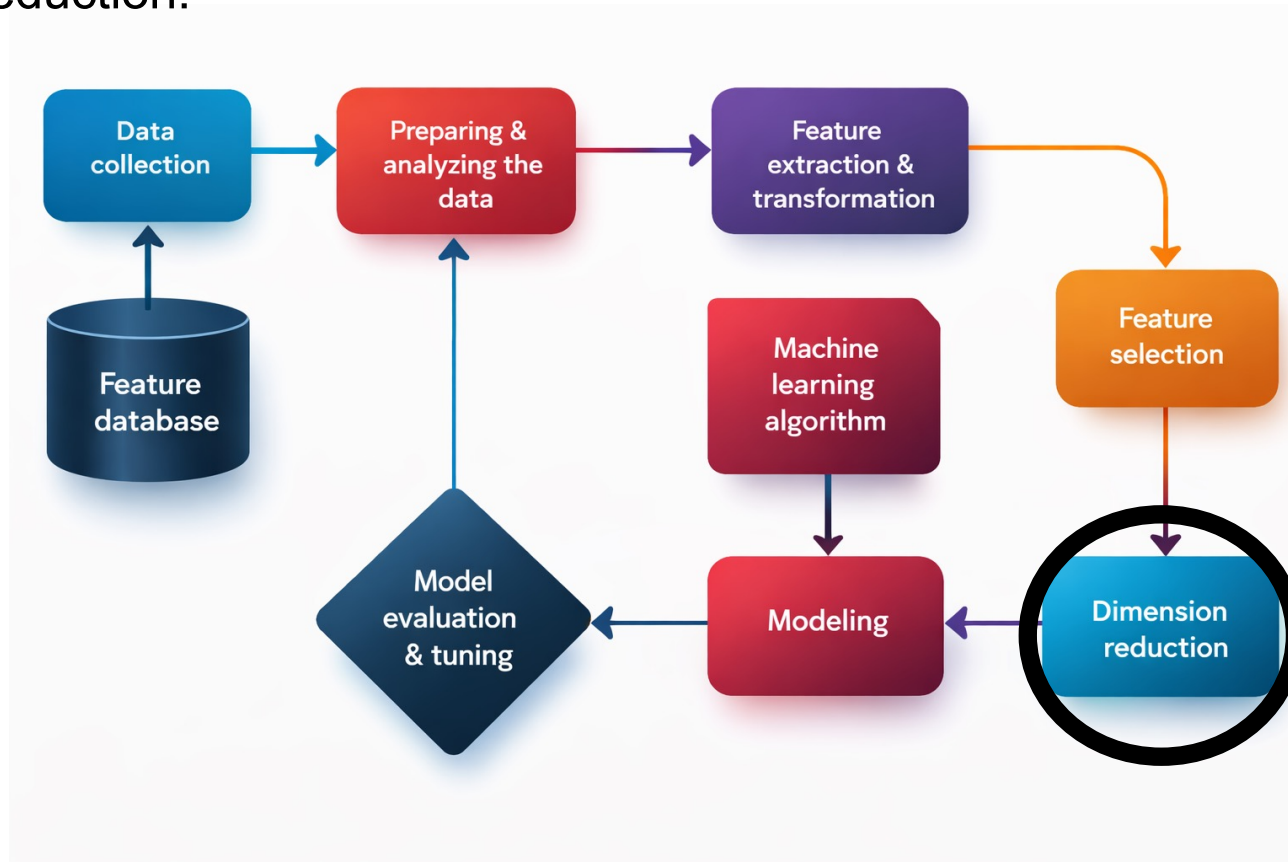


1. Digesting Measures: How does MLA decide?

Why do we need proper metrics?

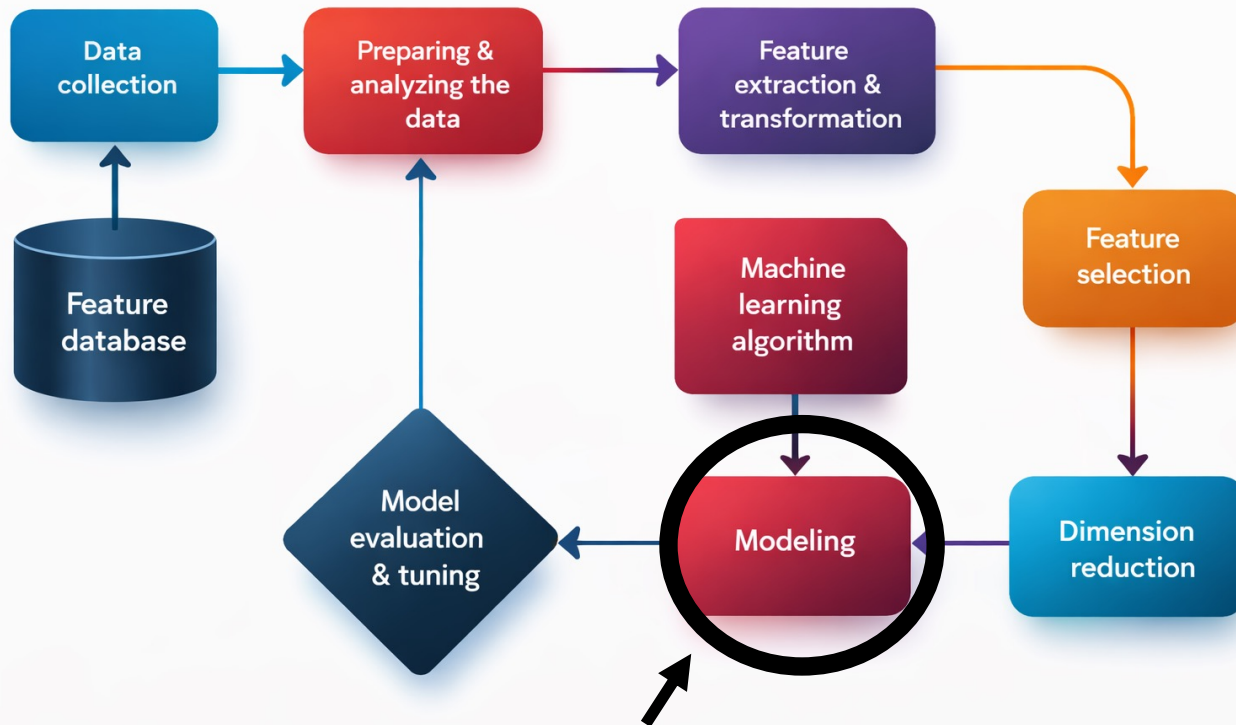
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- Dimension reduction:

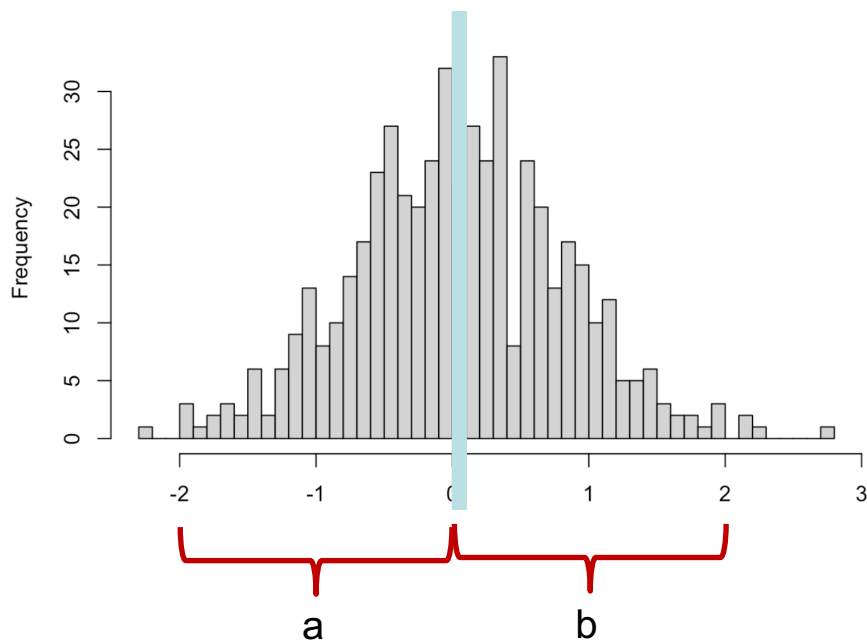


PCA:
calculation
of covariance,
variance
depend
on the mean

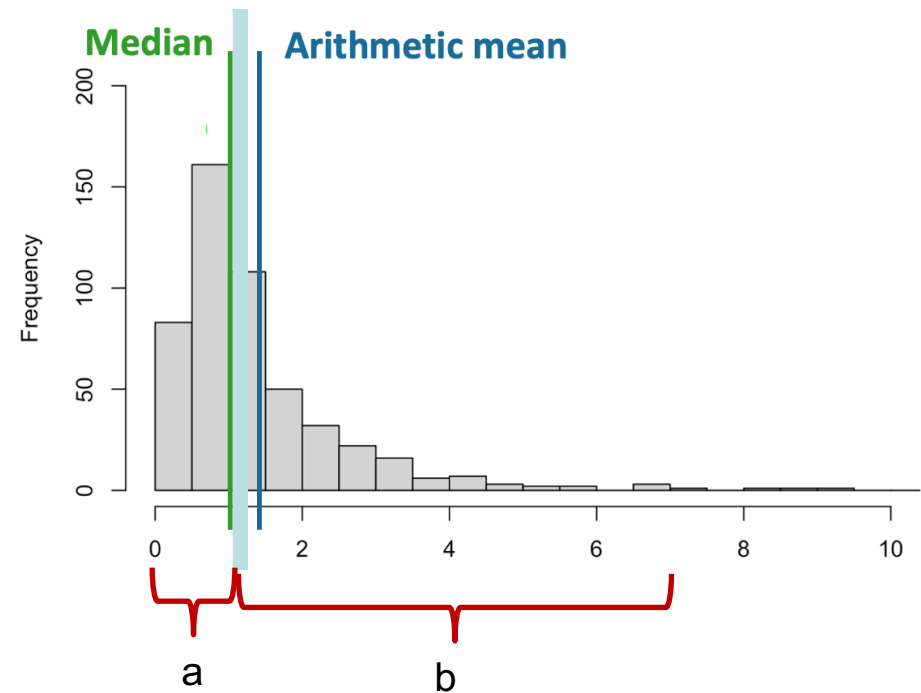
Why do we need proper metrics?



A lot of algorithms are **distance based**, e.g. KNN, SVM, k-means, ...
→ We need **correct metrics**!



Symmetric distributions:
Euclidian distances make sense
 $a = b$



Non-symmetric distributions:
Euclidian distances do not make sense
 $a \neq b$

Distance based algorithm example I: KNN

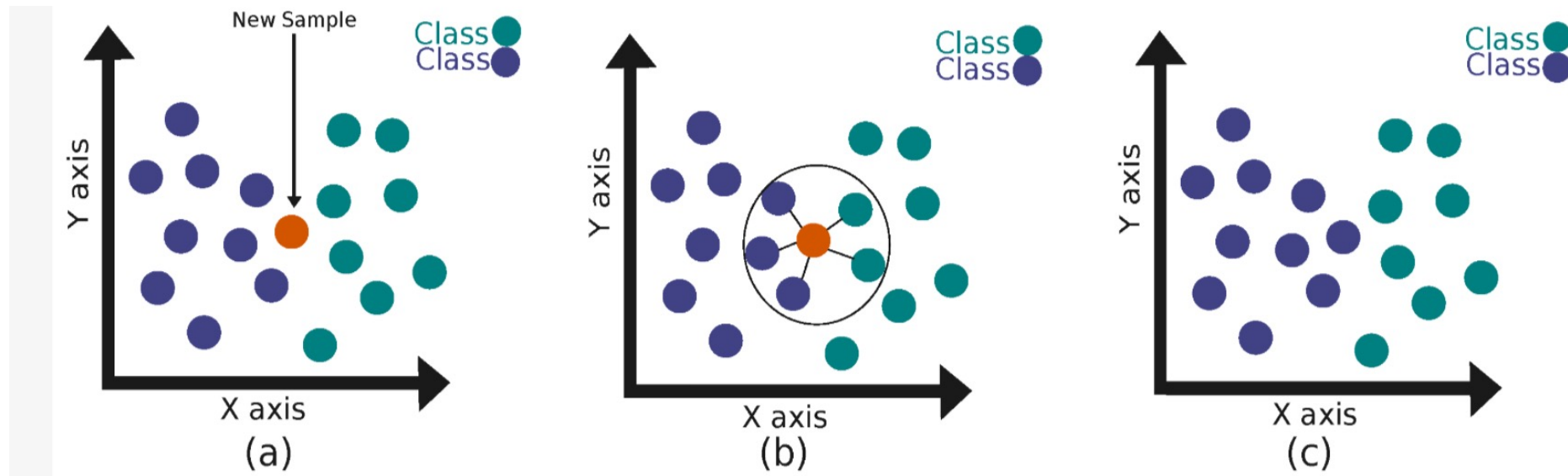
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K-NN with K=5

Orange: New Sample /
Unknown instance

Find k nearest
neighbors

Which class
predominates?



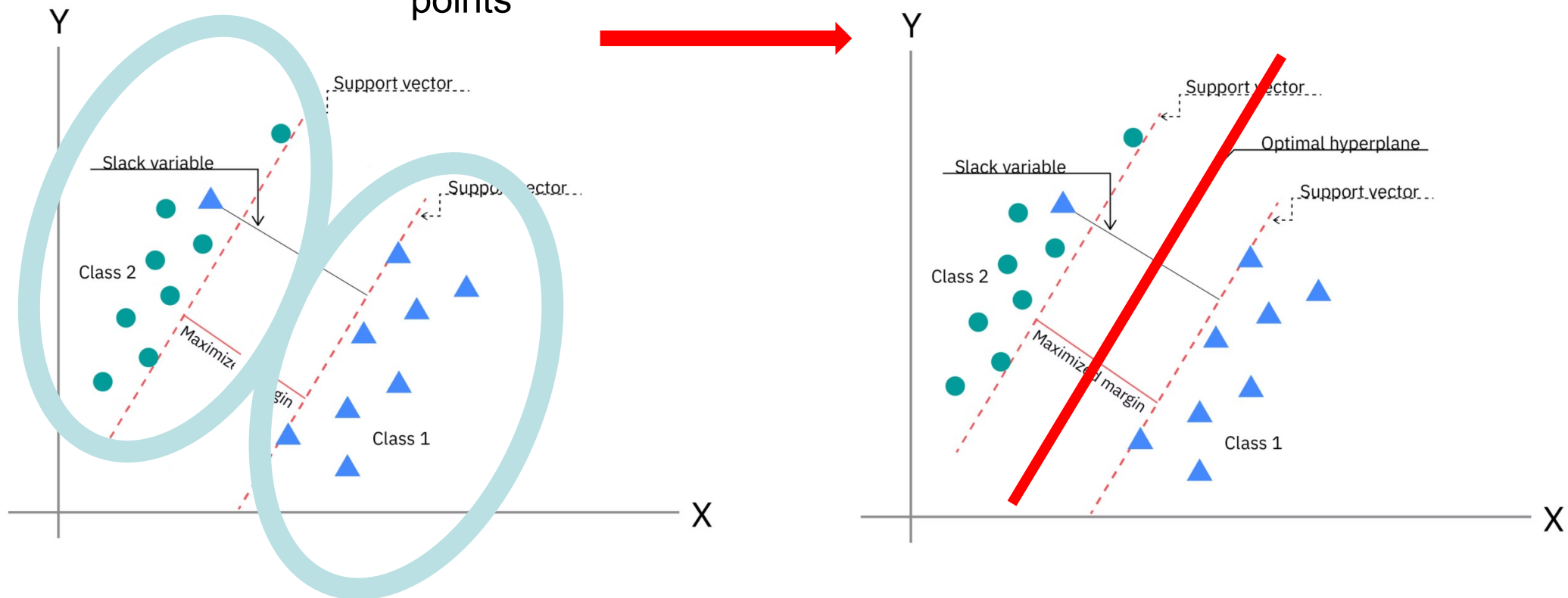
Maldonado-Romo, A., Montiel-Pérez, J. Y., Onofre, V., Maldonado-Romo, J., & Sossa-Azueta, J. H. (2024). Quantum K-Nearest Neighbors: Utilizing QRAM and SWAP-Test Techniques for Enhanced Performance. *Mathematics*, 12(12), 1872. <https://doi.org/10.3390/math12121872>

1. Digesting Measures: How does MLA decide?

Distance based algorithm example II: SVM

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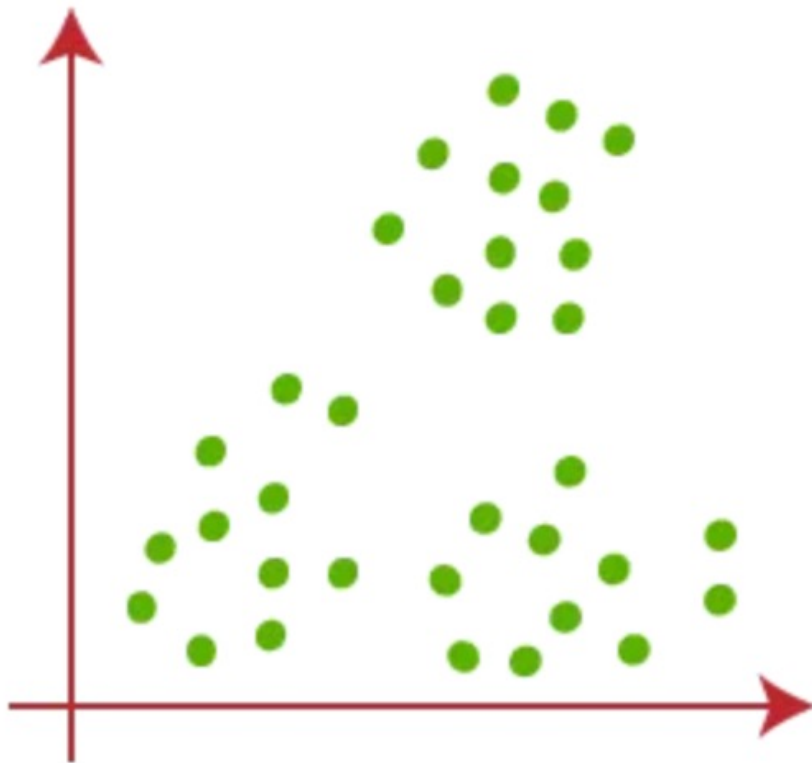
Find the line separating the data set that has the **maximal distance** to the data points



1. Digesting Measures: How does MLA decide?

Distance based algorithm example III: k-means¹⁷

Before K-Means

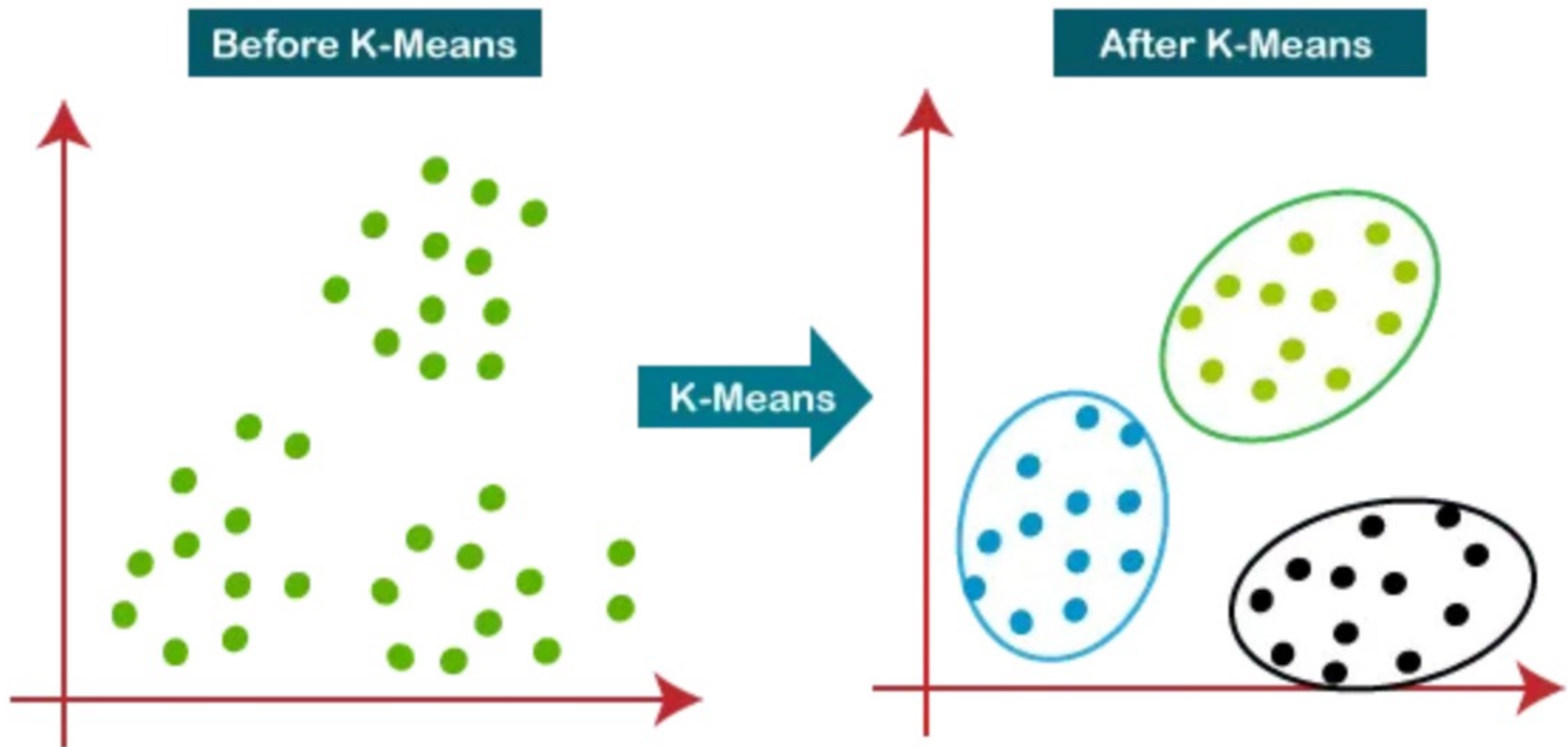


1.Specify K and Initialize: Choose the number of clusters K and randomly select KK initial cluster centers
Assign Points to Clusters: For each data point, **calculate its distance** to all centroids and assign it to the nearest one, forming temporary clusters.

2.Update Centroids: Recalculate each cluster center as the **mean (average)** of all points assigned to its cluster.

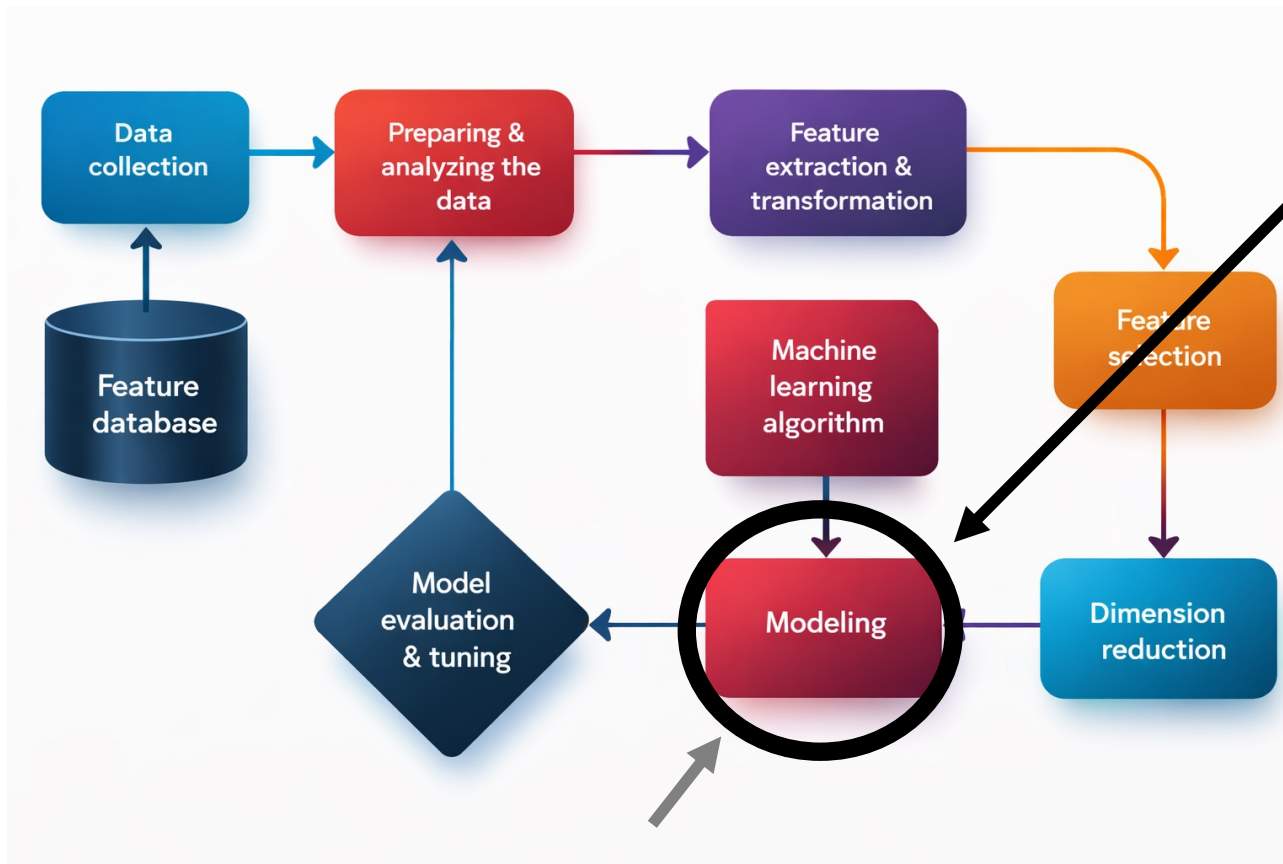
3.Repeat or Converge: Iterate steps 2 and 3 until centroids stabilize (change less than a threshold), assignments no longer change, or a maximum iteration limit is reached

Distance based algorithm example III: k-means¹⁸



1. Digesting Measures: How does MLA decide?

Why do we need proper metrics?

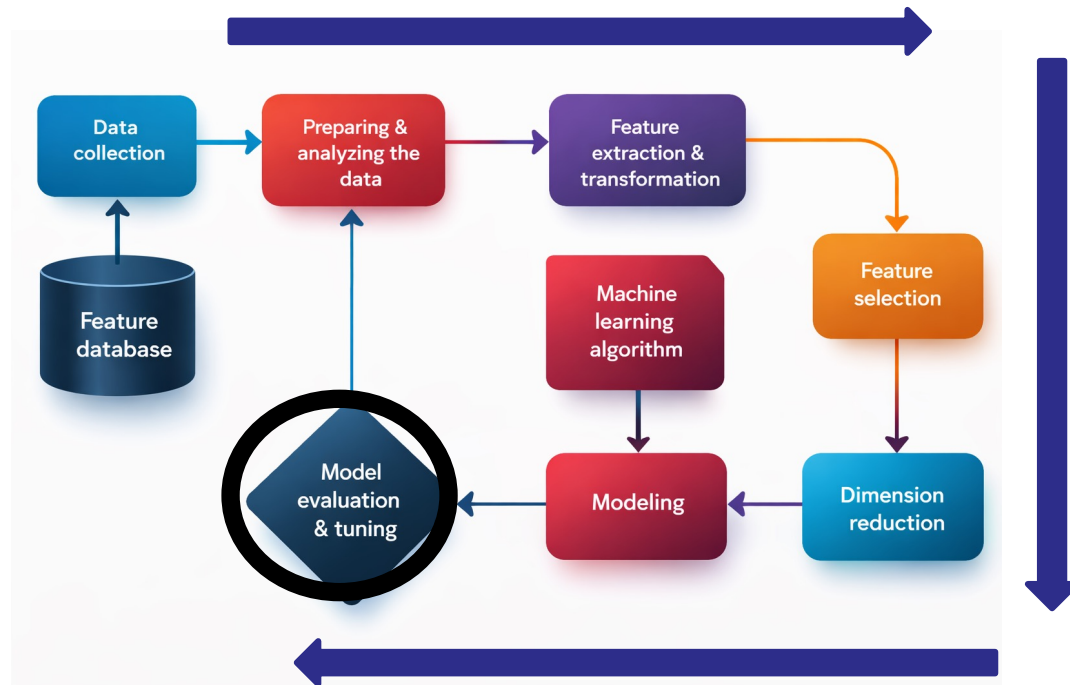


- But also...
 - **Regression (based)** algorithms need normality (e.g. NN)
 - **Gradient** based algorithms
 - **proper coordinate systems are needed!**

A lot of algorithms are **distance based**, e.g. KNN, SVM, k-means, ...
→ We need **correct metrics!**

Why do we need proper metrics

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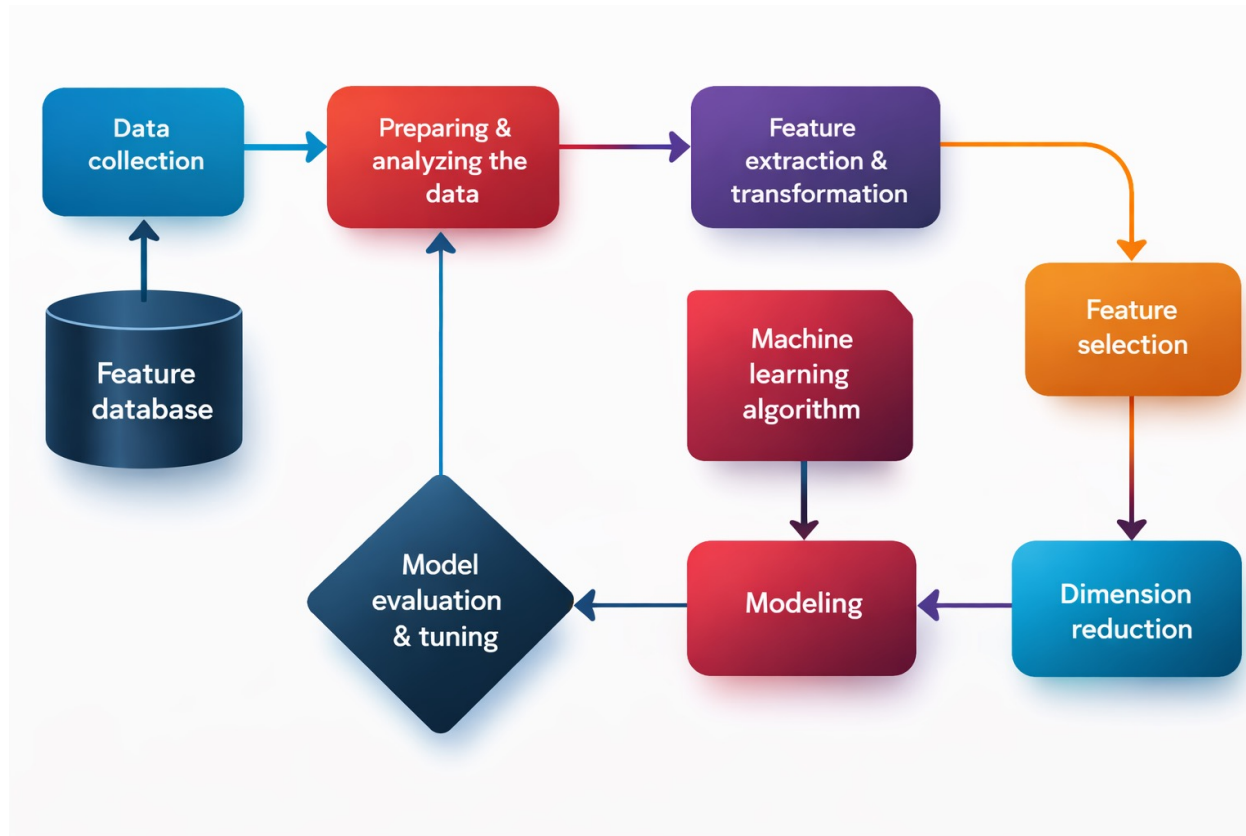
Neglecting Feature
Scale
Transformations
(FST)s
lead to unreliable
And lower scores

- Wrongly applied models (e.g. that assume normal distributions and/or are based on metrics
 - Non-normally distributed residuals
 - Accuracy scores become meaningless

Why do we need proper metrics?

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We need FSTs for every stage of the data evaluation process!

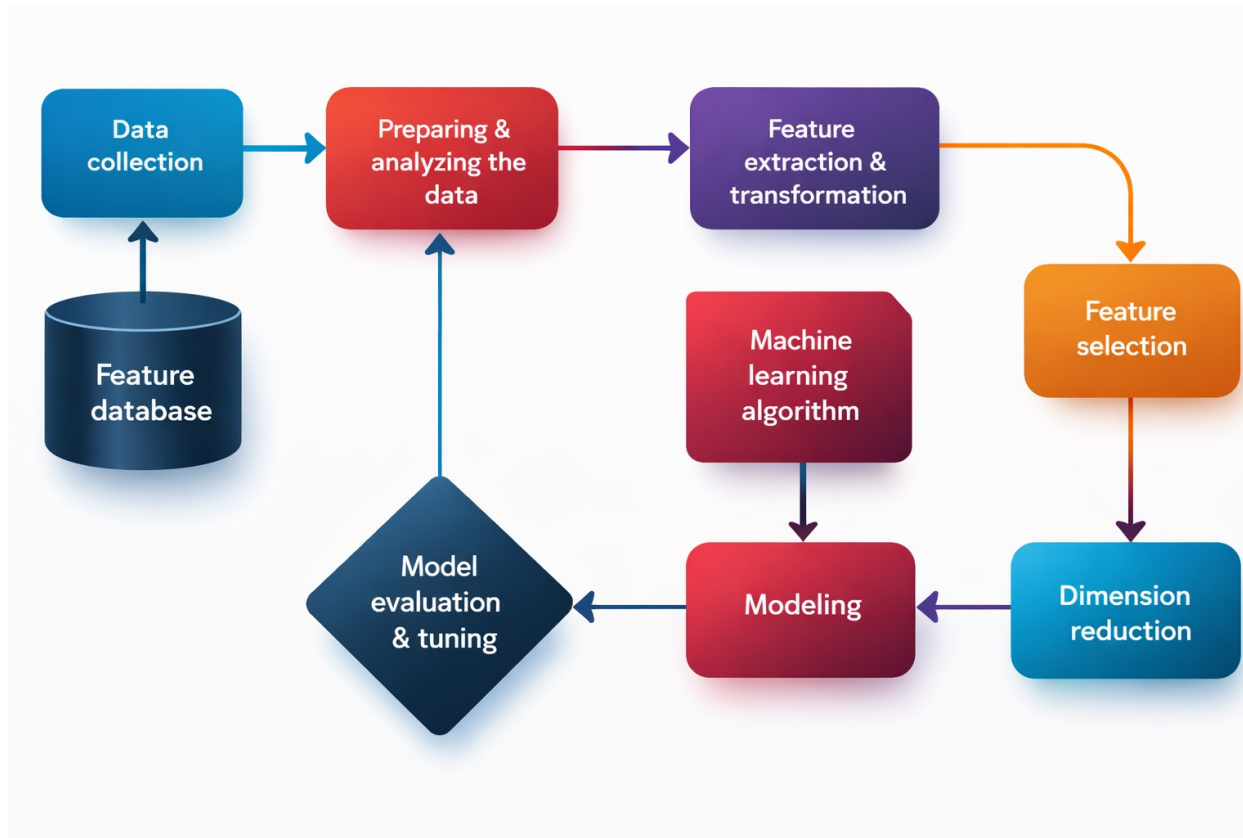


1. Digesting Measures: How does MLA decide?

Why do we need proper metrics?

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We need FSTs for every stage of the data evaluation process!



How can we transform the data?

1. Digesting Measures: How does MLA decide?

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- Providing **meaningful data reports** is **crucial for creating confidence** in your scientific outcome!
- The majority of **ML-Approaches are making decisions** based on
 1. distances (e.g. confidence intervals, k-means)
 2. angles (correlation, covariances, regression, gradients, etc.)
- Data spaces should have **Euclidean properties** (orthonormal base)
- **Symmetry or normality** is often an **undeniable assumption**!
- **Disobeyance** of mathematical, physical, or statistical requirements may **obscure our outcome**!

PS: We will neither discuss exceptions (e.g., entropy-based ML) nor cover all needs for compositional data!

Let us concentrate on basic recipes for reliable ML applications!

Our playground: Data Types & Measures

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Scale	Measure	Examples	Similarity measure	Central tendency	Dispersion measure
Binomial	success or failure	Rainy days	incomparable, equality	$n \cdot p$	$np(1 - p)$
(Multi-) nominal	memberships of categories	Soil types	membership, equality	mode, $\bar{x}_{geo}$, $n \cdot p_i$	entropy
Ordinal	as nominal plus < "is greater"	hurricane category	membership, rank order	median Q_{50}	$IQR = Q_{75} - Q_{25}$
Interval	distances $d(x_1, x_2) = x_1 - x_2 $	time, location	distance close to 0 , additive	<i>arithmetic mean:</i> $\bar{x}_{ar}$	<i>arithmetic stand. dev.:</i> s_{ar}
Rational	ratio: $t = x_1/x_2$	length, weight, isotope ratio	ratio close to 1, multiplicative	<i>geometric mean:</i> $\bar{x}_{geo}$	<i>geometric stand. dev.:</i> s_{geo}
Logistic	relative position between two limits: $\left(\frac{x-l}{u-l}\right)$	elevation, depth	log-ratio-distance, multiplicative	logistic mean $\bar{x}_{lgt}$	logistic variance s_{lgt}^2
Concentration	part of a constant sum	TOC, recovery rate	Aitchison distance, multiplicative	<i>geometric mean:</i> $\bar{x}_{geo}$	compositional variance

2. Data types, measures and moments

Two worlds of similarity

World of Σ

(Greek letter “sigma” = S
→ sum of discrete numbers/objects)

$$\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n$$

Common application:

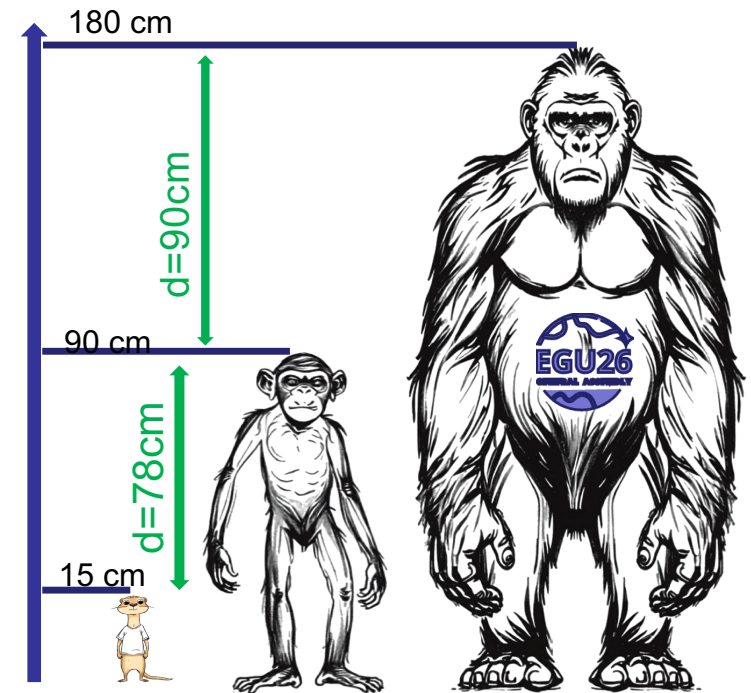
- a and b are the sizes of two object.
- How large is their difference d?

Imagination:

- distance (in amount, length, distance, etc.)

Interpretation

- > the closer the two value, the more similar they are.



Two worlds of similarity

World of Σ

Native inhabitants:

signed integer

$x \in \mathbb{Z} = \{a, -a: a \in \mathbb{N}_0 \wedge a + (-a) = 0\}$ for discrete scales

rational number

$x \in \mathbb{Q} = \{\frac{a}{b}, a, b \in \mathbb{Z} \wedge b \neq 0\}$ for continuous scales

real number

$x \in \mathbb{R}$: as rational numbers plus irrational numbers

Properties:

- **0** is the **center** (origin) and **neutral element** ($a+0=a$ resp. $a-a=0$)
- **Distances** $|a - b| > 0, a, b \in \mathbb{Q}/\mathbb{R}$ resp.
- $\|\vec{a} - \vec{b}\| > 0, \vec{a}, \vec{b} \in \mathbb{Q}^n/\mathbb{R}^n$ are **meaningful measures of similarity**



Operations $+$, $-$, $*$, $/$ can be applied to all inhabitants without leaving the range of the scale!



→ Σ is symmetric: $|a - b| = |b - a|$



$\min(\Sigma) = -\infty \wedge \max(\Sigma) = \infty$: Home of Normality $\mathcal{N}(\mu, \sigma)$

2. Meaningful measures

2.2. Σ and Π : Two different worlds of similarity

World of Π

(Greek letter “pi” = π)

→ product of discrete numbers/objects)

Common application:

a and b are the sizes of two object.

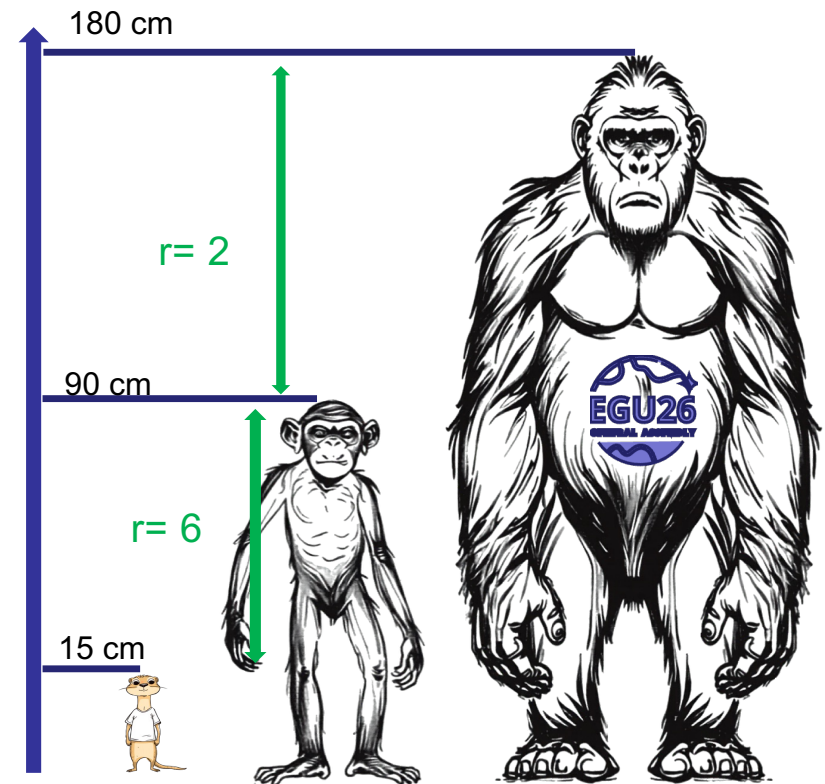
What is their relation (factor, ratio) r?

Thus, we calculate: $a * r = b$ or $b : a = b * a^{-1} = r$.

→ two complementary operators $*$ and $:$ ($\div$)

Interpretation

→ the closer the ratio to 1, the more similar are objects!



Two different worlds of similarity

World of Π

$$\sum_{i=1}^n x_i = x_1 + x_2 + \dots + x_n$$

Native inhabitants:

integer $x \in \mathbb{N} = \{1, 2, 3, \dots\}$ for discrete scales

positive rational number $x \in \mathbb{Q}_+ = \{\frac{a}{b} : a, b \neq 0\}$ for continuous scales

positive real number $x \in \mathbb{R}_+$

Properties:

- **1** is the **center** (origin) and **neutral element** ($a * 1 = a$ resp. $a * a^{-1} = 1$)
- **Ratios** $\frac{a}{b} > 0$, $a, b \in \mathbb{Q}_+ \vee \mathbb{R}_+$ **are meaningful measures of similarity**



only multiplication /division */ can be applied to all inhabitants without leaving the range of the scale!



Π is asymmetric: $\forall a > 1 \quad \exists a^{-1} \in]0, 1[_{\mathbb{R}}$

$$\Rightarrow |a| \neq |a^{-1}| = \left| \frac{1}{a} \right|; \quad \text{e.g. } \left| \frac{1}{2} \right| \neq \left| \frac{2}{1} \right|$$

Arithmetic mean is tied to full real space!

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Let's look at the arithmetic mean:

$$\bar{x}_a = \frac{1}{n} \sum_{i=1}^n x_i$$

$$n \cdot \bar{x}_a = \sum_{i=1}^n x_i$$

$$0 = \left(\sum_{i=1}^n x_i \right) - (\bar{x}_a + \bar{x}_a + \dots + \bar{x}_a)$$

$$0 = \sum_{i=1}^n (x_i - \bar{x}_a)$$

$$\sum_{i: x_i < \bar{x}} (x_i - \bar{x}_a) = - \sum_{i: x_i > \bar{x}} (x_i - \bar{x}_a)$$

→ The arithmetic mean is meaningful in a symmetrical feature space, only!

$$\Sigma : \mathbb{R}, \mathbb{Q}(\mathbb{Z})$$

Sums and differences may result in
negative amounts!

Thus, Σ calls for **full real/rational space!**

The **most popular representatives** are the
arithmetic mean and the arithmetic **standard deviation**.

→ They **pass over** their properties and
limitations to the **absolute majority of
statistical methods!**

Σ is the **home for normality** which requires
free vision to $\pm\infty$

Predominant home for
statistical methods and
Textbooks about Stats!

$$\Pi : \mathbb{R}^+, \mathbb{Q}^+ (, \mathbb{N})$$

Positive numbers cannot leave the **positive
branch** by multiplication/division!

Π has only a few representatives but cares for
the **absolute majority of real-world data!**

The implicit **skewness hamper linearity!**

Π is the **home for log-normality**

Predominant home for
real-world data and
empirical Sciences!

Ways out of the mess!

$\Sigma : \mathbb{R}, \mathbb{Q}$

Home of **methods**

$\Pi : \mathbb{R}^+, \mathbb{Q}^+, \mathbb{N}$

Home of real-world **data**



Let us start from real world multiplicative (rational) feature space:

a: **1** is the **center** and the **neutral element** of the rational scale!

b: **the closer** the ratio to 1, **the more** similar the objects are!

but:

→ Π is asymmetric:

for two measure $0 < a < b$: $\left| \frac{a}{b} \right| < \left| \frac{b}{a} \right|$

but: $\left| \log \left(\frac{a}{b} \right) \right| = \left| \log \left(\frac{b}{a} \right) \right|$

because: $\left| \log \left(\frac{a}{b} \right) \right| = |\log a - \log b| = |\log b - \log a| = \left| \log \left(\frac{b}{a} \right) \right|$



The log-ratio of two measures is a symmetric measure!



The ratio is meaningful, but bends the feature space!

The log-ratio of two measures is a symmetric measure!

Therefore, the **arithmetic mean of log-transformed rational data** should be meaningful as well:

$$x \rightarrow x' = \log(x): \quad \bar{x}'_{\text{ar}} = \frac{1}{n} \sum \log x$$

But, a log-mean is **difficult to imagine**. Let us transform the log-mean back:

$$x' \rightarrow x = e^{x'}: \quad e^{\bar{x}'_{\text{ar}}} = e^{\frac{1}{n} \sum \log x_i} = \left(\prod_{i=1}^n e^{\log x_i} \right)^{\frac{1}{n}} = \sqrt[n]{x_1 \cdot x_2 \dots x_n} = \bar{x}_{\text{geo}}$$



The geometric mean is meaningful on multiplicative (ratio) feature scales!

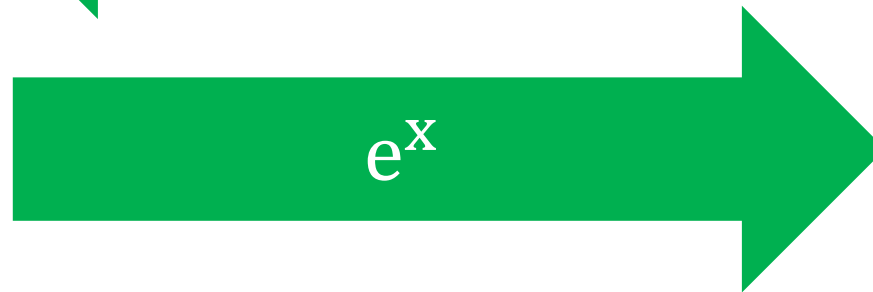
Ways out of the mess!

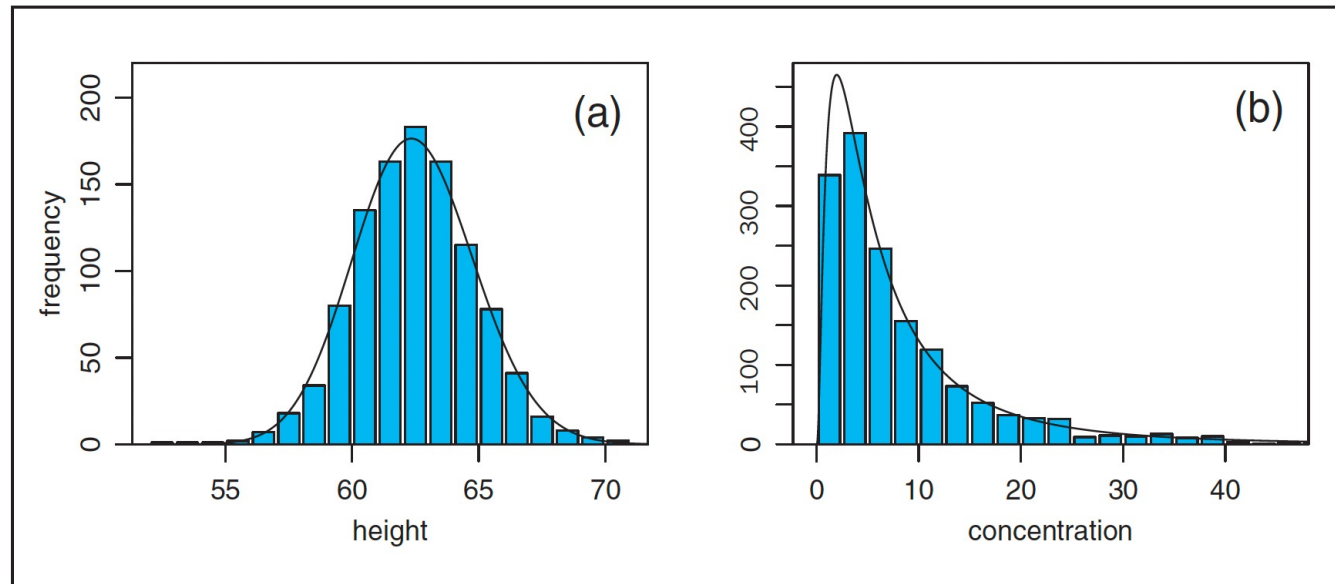
$\Sigma : \mathbb{R}, \mathbb{Q}$

Home of **methods**

$\Pi : \mathbb{R}^+, \mathbb{Q}^+, \mathbb{N}$

Home of real-world **data**





Distribution fits the Normal with
 $p = 0.75$ and the log-Normal
with $p = 0.74$

Distribution fits the log-Normal
with $p = 0.41$
but not the Normal ($p < 0.0001$)



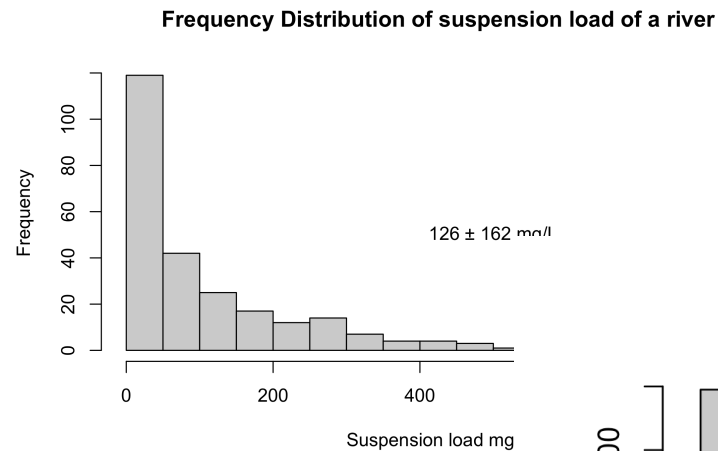
Takeaway:

- Assuming the log-Normal doesn't hurt, but the Normal!
- You can infinitely flatten a curve, but never bend a straight line!

Limpert, E., Stahel, W.A., Abbt, M., 2001. Log-normal Distributions across the Sciences: Keys and Clues. BioScience 51, 341. [https://doi.org/10.1641/0006-3568\(2001\)051%255B0341:LNDATS%255D2.0.CO;2](https://doi.org/10.1641/0006-3568(2001)051%255B0341:LNDATS%255D2.0.CO;2)

1- σ for geometric mean

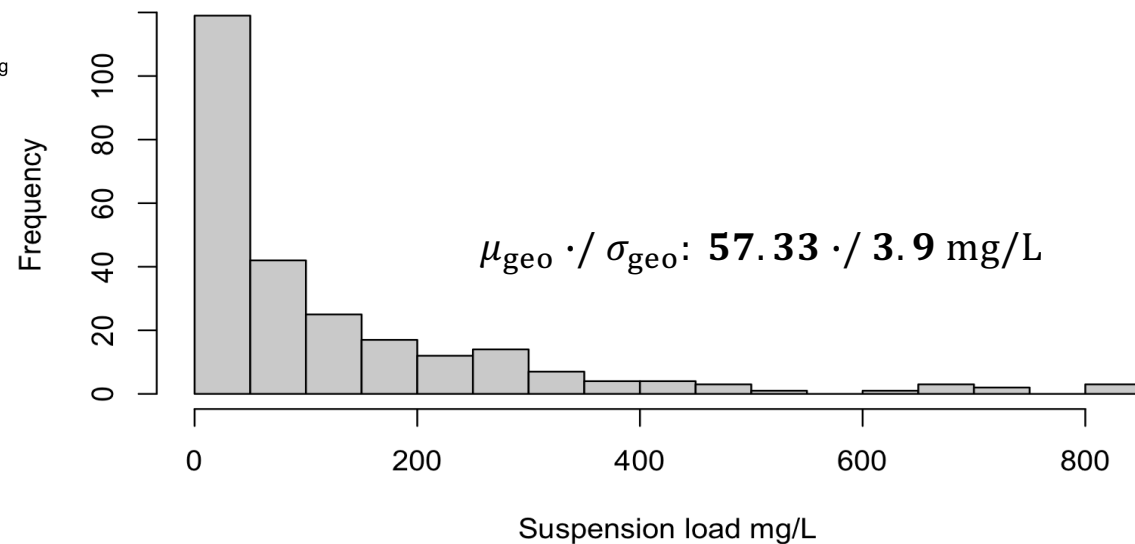
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Recipe:

1. transform your data: $\log(x)$
2. analyse your data
3. transform desired parameters back: e^x

Frequency Distribution of suspension load of a river



But, keep in mind:
 $\mathbb{R}^+$ is multiplicative

2. Meaningful measures

3.0. Normal or log-Normal?

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Property	Normal distribution (Gaussian, or additive normal, distribution)	Log-normal distribution (Multiplicative normal distribution)
Effects (central limit theorem)	Additive	Multiplicative
Shape of distribution	Symmetrical	Skewed
Models		
Triangle shape	Isosceles	Scalene
Effects at each decision point x	$x \pm c$	x^x / c'
Characterization		
Mean	$\bar{x}$, Arithmetic	$\bar{x}^*$, Geometric
Standard deviation	s, Additive	s^* , Multiplicative
Measure of dispersion	$cv = s/\bar{x}$	s^*
Interval of confidence		
68.3%	$\bar{x} \pm s$	$\bar{x}^* \times / s^*$
95.5%	$\bar{x} \pm 2s$	$\bar{x}^* \times / (s^*)^2$
99.7%	$\bar{x} \pm 3s$	$\bar{x}^* \times / (s^*)^3$
Notes: cv = coefficient of variation; $\times /$ = times/divide, corresponding to plus/minus for the established sign $\pm$.		

Is this brand-new science? Definitely not:

Mean, Normal-Distribution and CLT published by C-F Gauss (1809) and Geometric Mean, Sd and CLT for Log-Normal D. McAllister 1879;

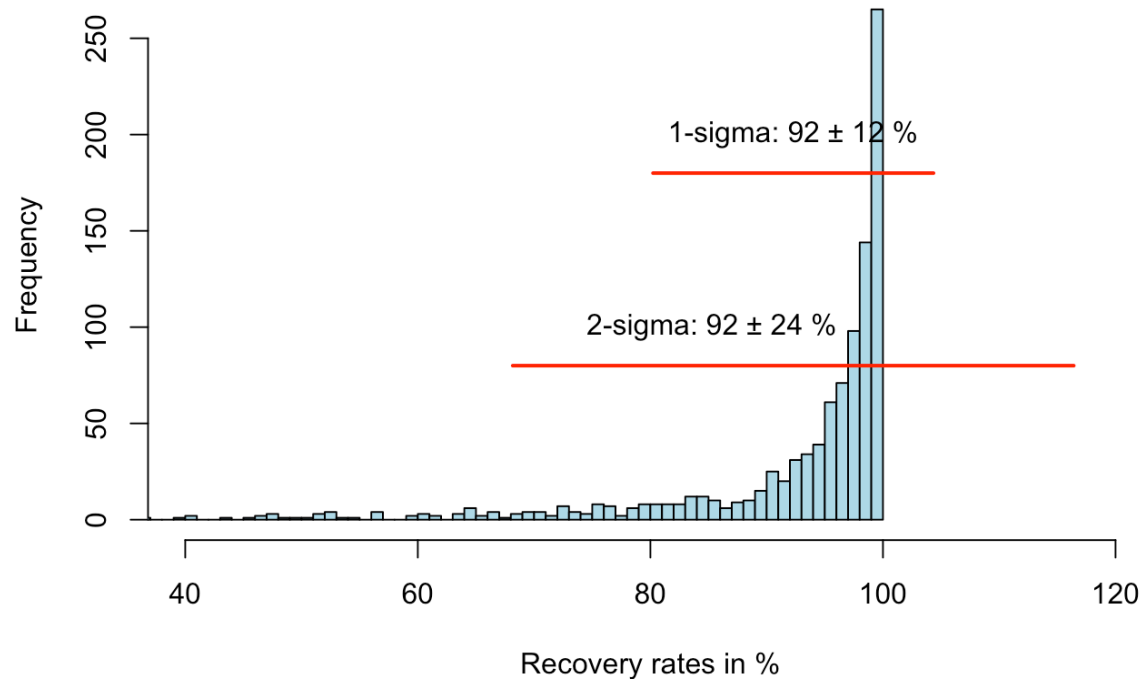
The Relation between both has been reported by F. Galton 1989

Limpert, E., Stahel, W.A., Abbt, M., 2001. Log-normal Distributions across the Sciences: Keys and Clues. BioScience 51, 341.

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2. Meaningful measures

Percent area recovered within 30 years



Here we have a lower and an upper limit, which can not be exceeded in a meaningful way!

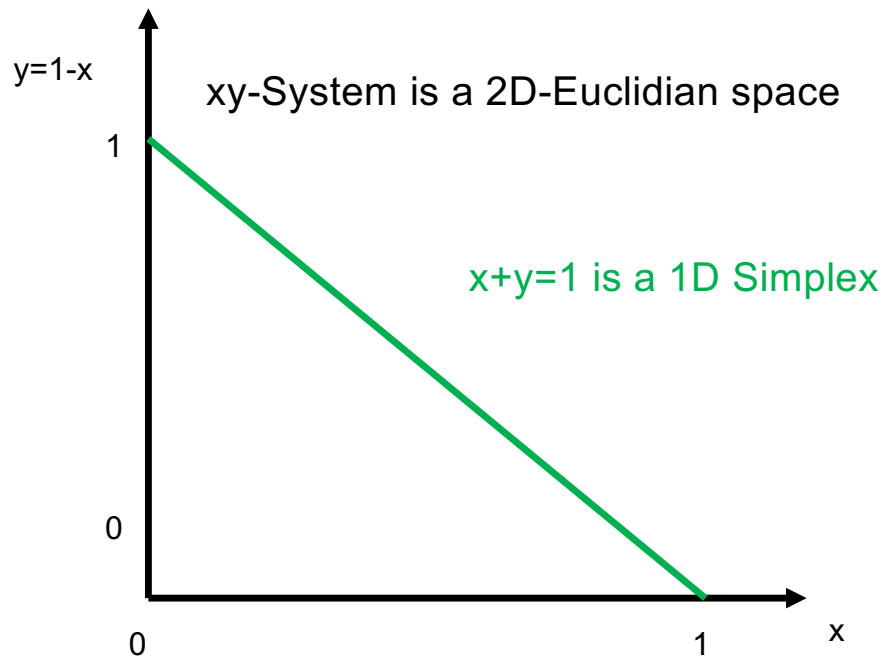
→ every value y has two information: distance to the left and to the **right boundary!**

Both informations are not independent!

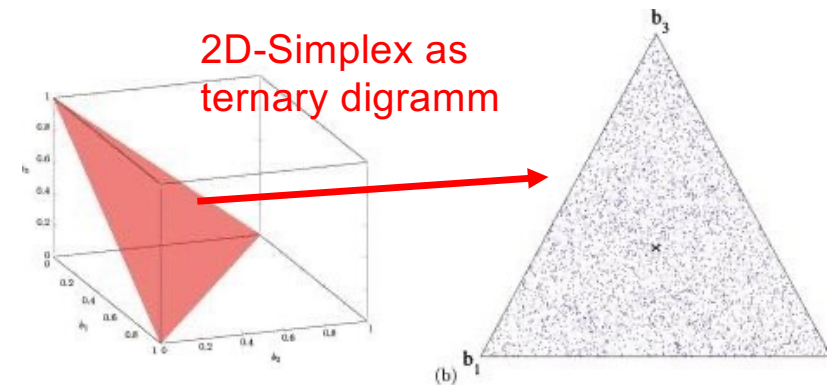


Double constraint Spaces are compositional by nature

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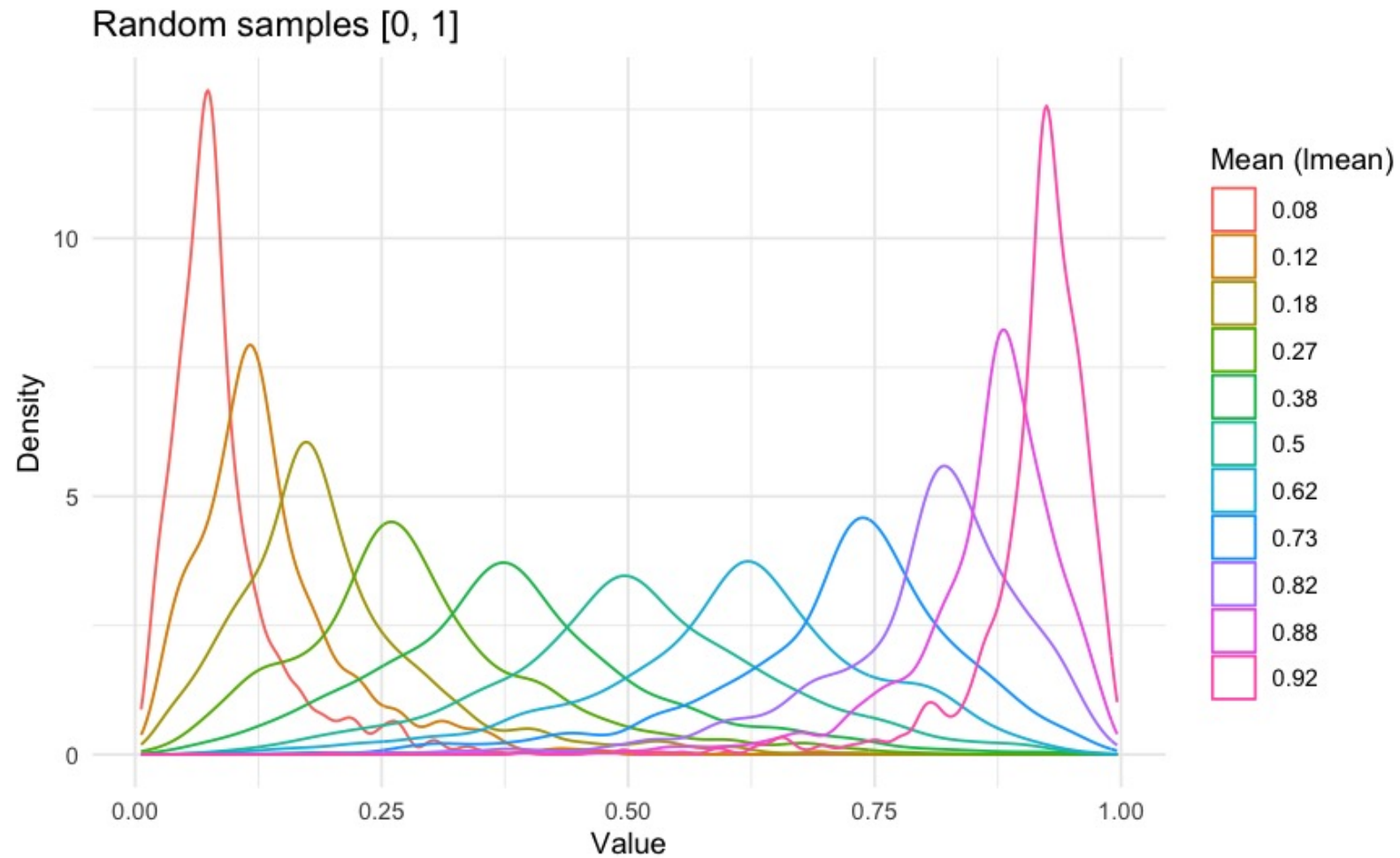


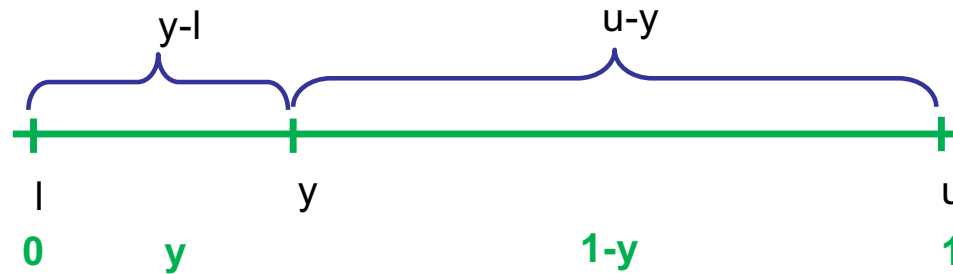
3D-Space with $x+y+z=1$



Feature spaces have an upper limit!
→ They are **constraint by a constant sum!**

2. Meaningful measures





But we can open our double-bounded feature scale by using a logratio transformation:

$$y' = \log\left(\frac{y-l}{u-y}\right)$$

$$y' = \log\left(\frac{y}{1-y}\right) : y \in]0,1[$$

Calculate your parameter p (= mean, variance, σ -intervals) and transform derived values back:

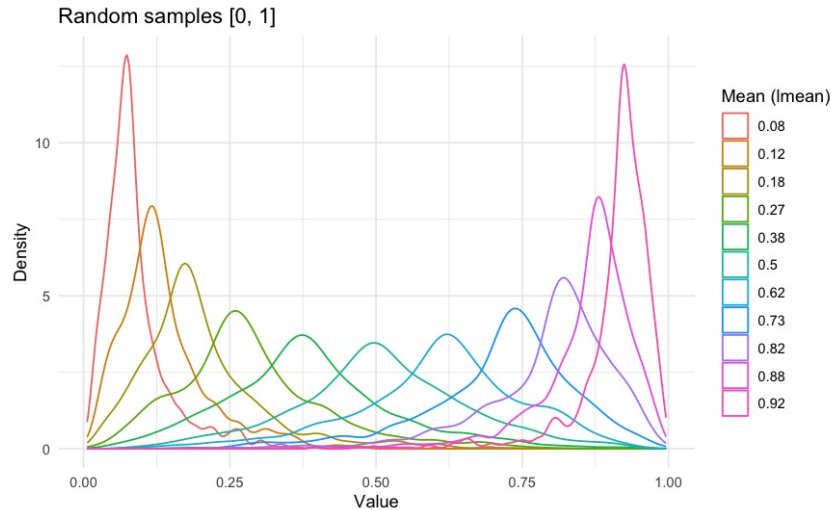
$$y = l + \frac{u-l \cdot e^p}{1 + e^p}$$

$$y = \frac{e^{y'}}{1 + e^{y'}} : y \in]0,1[[$$

Johnson, N.L., 1949. Systems of Frequency Curves Generated by Methods of Translation. *Biometrika* 36, 149–176.

The logistic moments

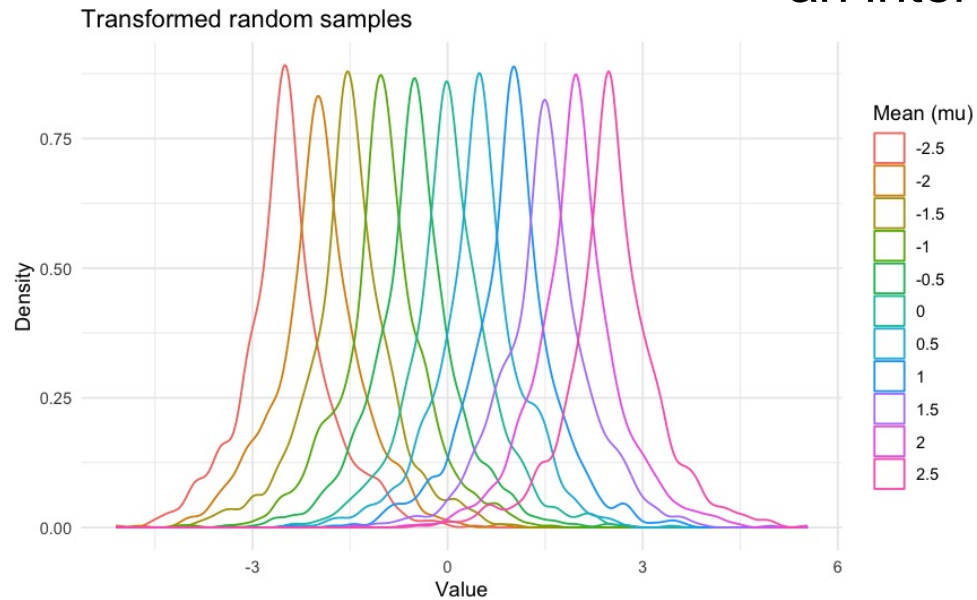
44



$$y \rightarrow y' \in \mathbb{R}$$

$$y' = \log\left(\frac{y}{1-y}\right)$$

full properties of
an interval scale



$$y = \frac{e^{y'}}{1 + e^{y'}}$$

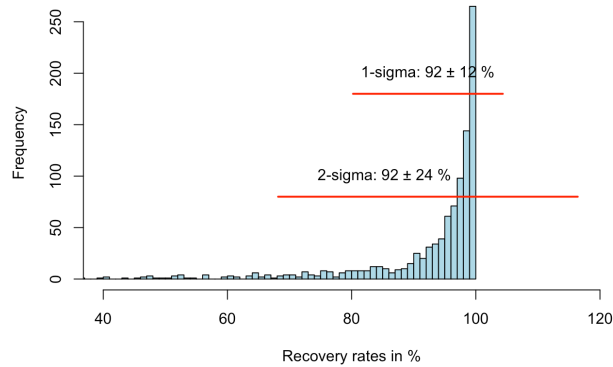
$$y' \rightarrow y \in]0, 1[\mathbb{R}_+$$

2. Meaningful measures

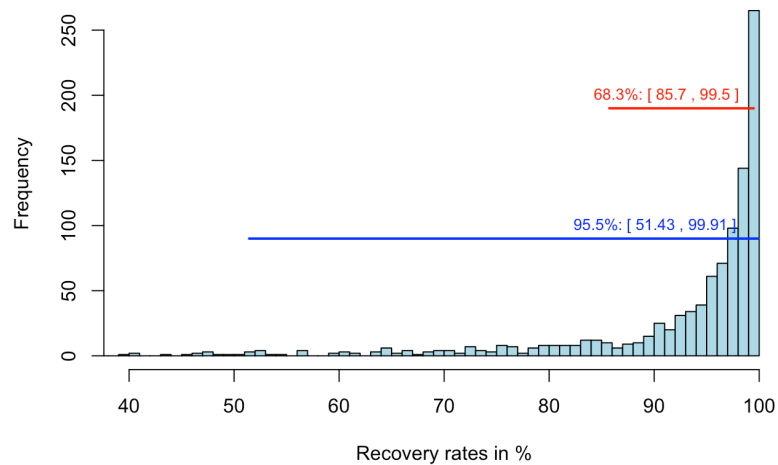
Meaningful intervals:

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Percent area recovered within 30 years

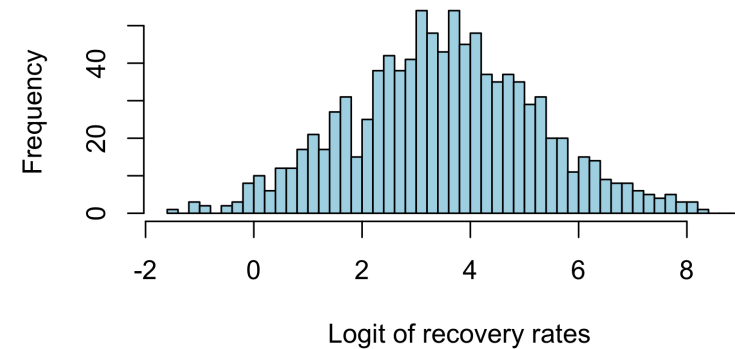


Percent area recovered within 30 years



$$y' = \log\left(\frac{y}{1-y}\right)$$

Logit-transformed data



$$y = \frac{e^{y'}}{1 + e^{y'}}$$

2. Meaningful measures

Take aways (FST)

- **Most real-world data** is only **meaningful** in $\mathbb{R}^+$
- $\mathbb{R}^+$ indicates **multiplicative similarity** measure
- $\mathbb{R}^+$ indicates **skewed distance** measure
- Most **statistical methods** assuming data living in full $\mathbb{R}$
- $\mathbb{R}$ provides **additive, symmetric distances** as **similarity measures**
- **log()** transforms $\mathbb{R}^+ \rightarrow \mathbb{R}$ and turns **multiplication into addition**
- **exp()** transforms $\mathbb{R} \rightarrow \mathbb{R}^+$ and turns **addition into multiplication**
- **disobeying mathematical rules** results in **senseless** informations
- a high amount of **real-world data** has an **upper limit u** in $\mathbb{R}^+$
- double constraint data (**DCD**) is naturally 2-dimensional (or 2n-D)
- **DCD** is multiplicative (ratios are meaningful)
- **log-ratio-** (logistic-) transformation **opens DCD to $\mathbb{R}$**
- Ignoring FS-constraints leads to **meaningless results**



Requirements of feature scales for correlation and linear regression:

- **Regressend or dependent variable** has to be **normal!**
- Predictor variables have to be at least **symmetric distributed**, ideally normal!
- all scales have to be **interval scales** resp. **additive unlimited!**
- arithmetic **mean** and **standard deviation** have to be **meaningful!**

Sounds challenging with real world data!

In particular, **double-constraint feature scales** have a **compositional nature**, indicating **spurious correlation** under the constant-sum constraint.

let us consider

$\vec{x} = ((x_1 - \bar{x}), (x_2 - \bar{x}), \dots, (x_n - \bar{x}))$ and
 $\vec{y} = ((y_1 - \bar{y}), (y_2 - \bar{y}), \dots, (y_n - \bar{y}))$ as 2 vectors $\mathbb{R}^n$

Then we may interpret the correlation coefficient as cosine of the angle between $\vec{x}$ and $\vec{y}$:

$$\cos \angle(\vec{x}, \vec{y}) = \frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\| \|\vec{y}\|} = \frac{\sum_{i=1}^n x_i y_i}{\sqrt{\sum_{i=1}^n x_i^2} \sqrt{\sum_{i=1}^n y_i^2}} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} = r$$

$\langle \vec{x}, \vec{y} \rangle$: Scalar product and $\|\vec{x}\|$ is the norm of the vector..

Extreme cases:

1. case: $r=0$

→ $\cos \angle(\vec{x}, \vec{y})=0 \rightarrow \angle(\vec{x}, \vec{y}) = 90^\circ$

→ both variable vectors are rectangular to each other!

2. case: $r=1$

→ $\cos \angle(\vec{x}, \vec{y})=1 \rightarrow \angle(\vec{x}, \vec{y}) = 0^\circ$

→ both variable vectors point to the same direction

3. Fall: $r=-1$

→ $\cos \angle(\vec{x}, \vec{y})=-1 \rightarrow \angle(\vec{x}, \vec{y}) = 180^\circ$

→ both variable vectors point to the same direction

Our data space require Euclidian properties: an orthonormal base!

(spurious correlations contradict these requirements!)

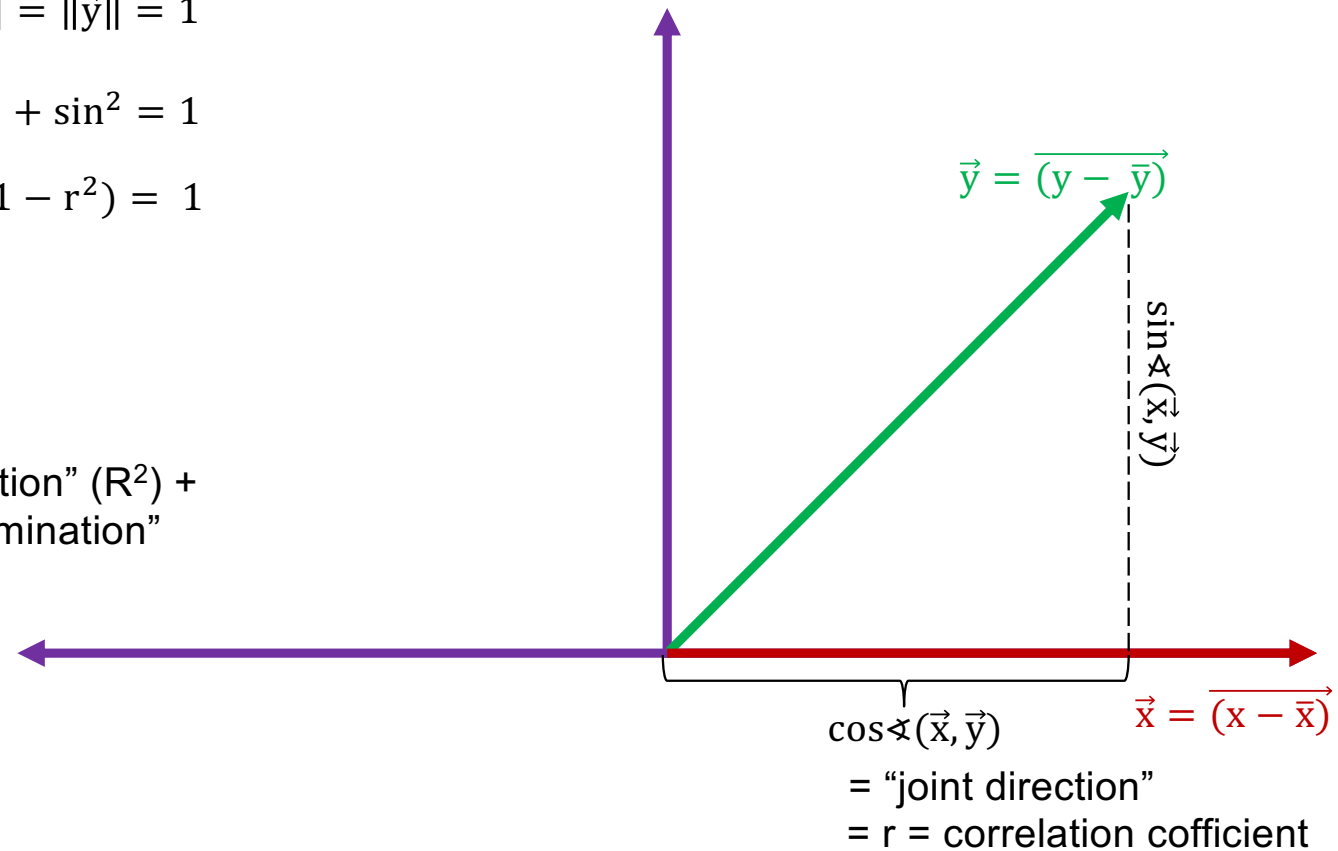
applying Pythagoras for $\|\vec{x}\| = \|\vec{y}\| = 1$

$$\cos^2 + \sin^2 = 1$$

$$r^2 + (1 - r^2) = 1$$

COD + COnD = 100%

“coefficient of determination” (R^2) +
“coefficient of non-determination”
= 100 %

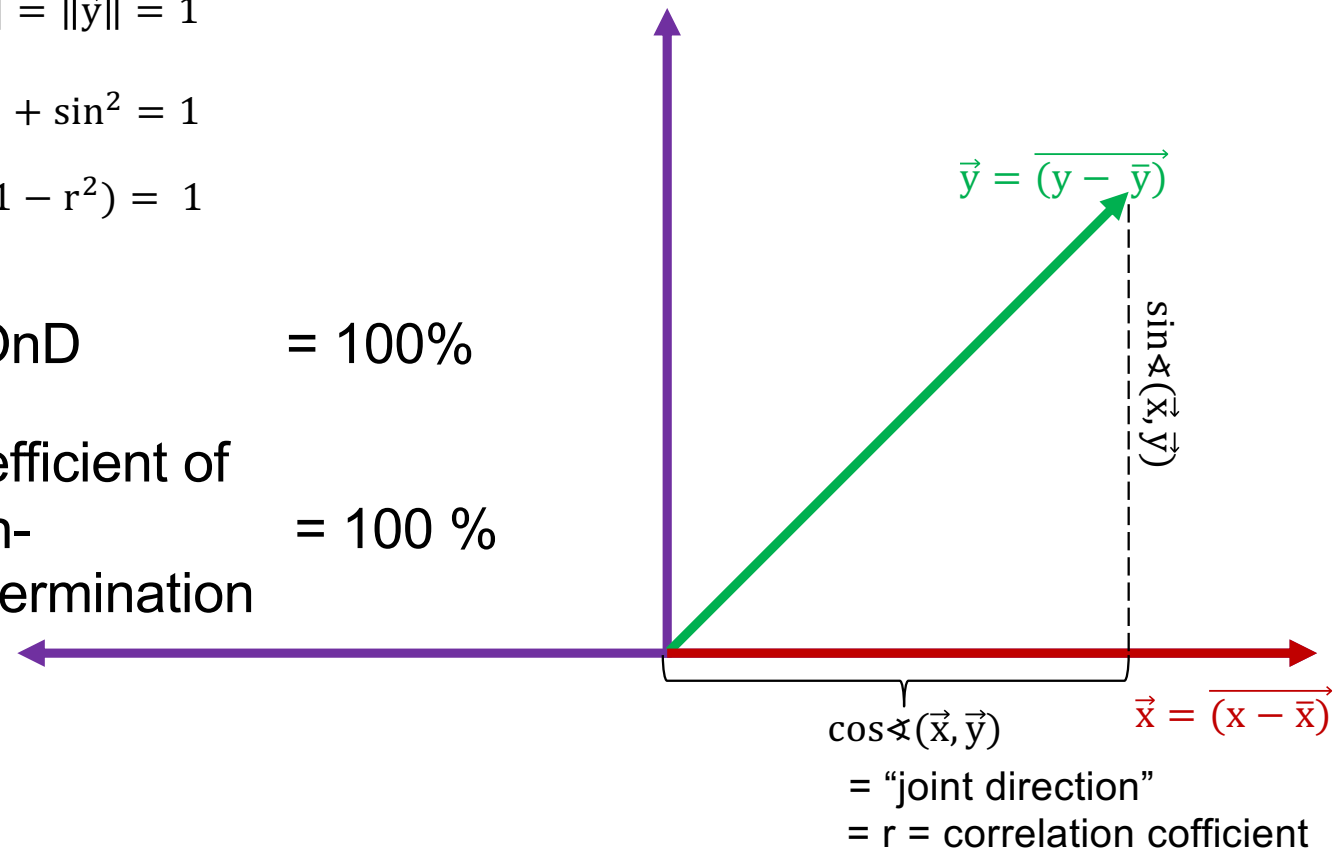


applying Pythagoras for $\|\vec{x}\| = \|\vec{y}\| = 1$

$$\cos^2 + \sin^2 = 1$$

$$r^2 + (1 - r^2) = 1$$

COD	+	COnD	= 100%
coefficient of determination” (R^2)	+	coefficient of non- determination	= 100 %



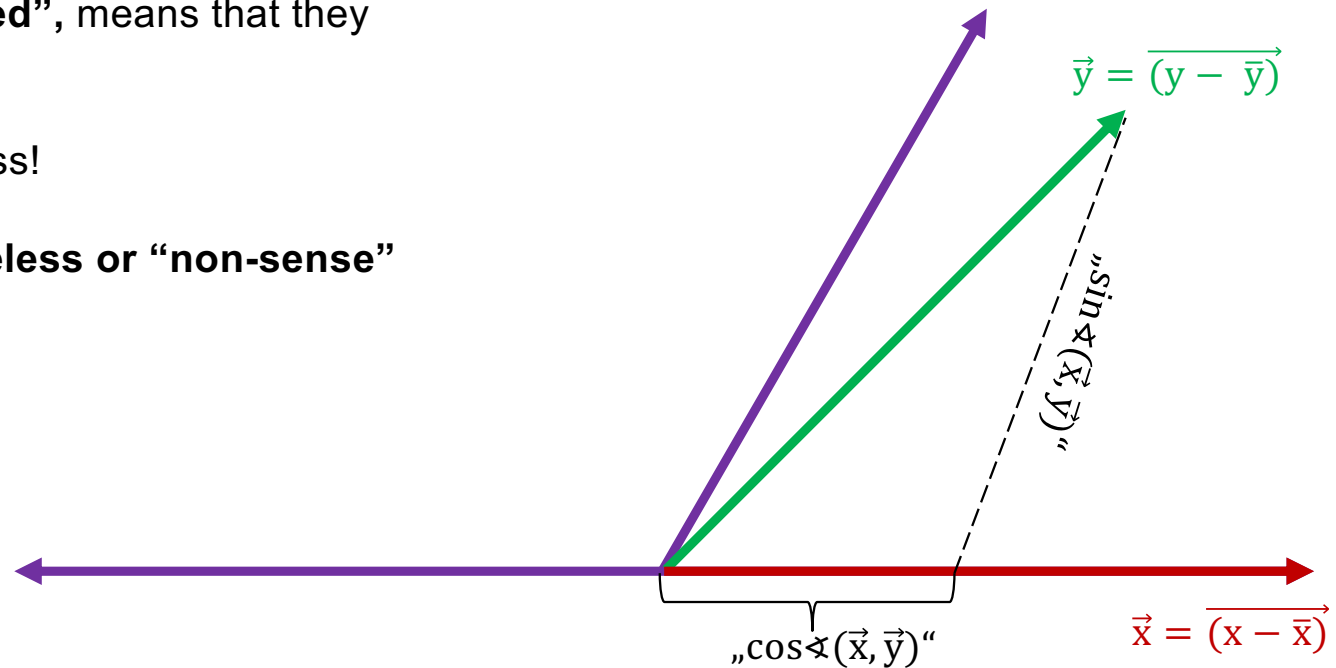
Correlation, Cosine, Scalar product

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If two variables are “**spurious correlated**”, means that they bases are **not rectangular**!

→ cos, correlation, COD are meaningless!

→ any calculated correlation is senseless or “non-sense” in sensu strictu!



Spurious correlation obscure the message in our data!

Spurious correlation produce artificial pattern!

Logratio-transformations are able to produce orthonormal bases (Euclidian metrics)



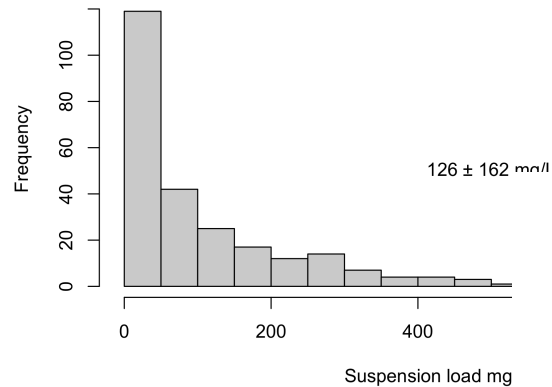
Outline

0. Motivation
1. Digesting Measures: How does MLA decide?
2. Meaningful measures: Truth beyond stats textbooks
- 3. Recipes: Preparing real-world data for ML**
4. Hands-on exercises

- **Step 1: Check the distribution** of the data using histograms, boxplots, or Q-Q plots to assess skewness and identify potential outliers. If the data has a lower limit greater than zero, shift the data by adding a constant to all values to ensure that all values are positive before applying the log-transformation.
- **Step 2:** Apply the **log-transformation** to the data using the formula $x' = \log(x)$, where x is the original data and x' is the transformed data.
- **Step 3:** After log-transformation, **check the distribution of the transformed data** to ensure that it is more symmetric and closer to a normal distribution.
- **Step 4: Calculate the arithmetic mean and standard deviation** of the log-transformed data, which are now appropriate measures of central tendency and variability for the transformed data.
- **Step 5: Back-transform the results** to the original scale by applying the **exponential function** $x = e^p$ to the mean ($p = \bar{x}_{\log(x)}$) and standard deviation ($p = s_{\log(x)}$) of the log-transformed data.
- After back transformation, the mean will be the **geometric mean** $\bar{x}_{geo}$ of the original data, and the **standard deviation** s_{geo} will be the geometric standard deviation of the original data.
- Due to the **direct multiplicative nature** of your original data, you can now report the 1-sigma range of the original data as the geometric mean $\times$ / geometric standard deviation: $\bar{x}_{geo} \times / s_{geo}$.
- ➔ Even applying higher sigma intervals or small sample sizes, these ranges cannot exceed $\mathbb{R}^+$ resp. $\mathbb{Q}^+$

Recipe 1: $0 < x$

Frequency Distribution of suspension load of a river

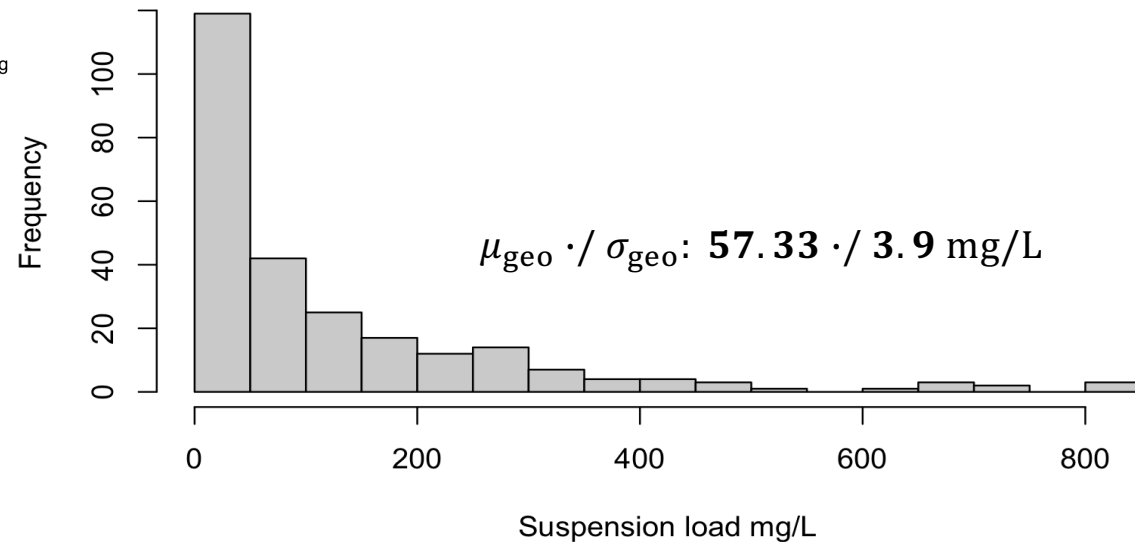


But, keep in mind:
 $\mathbb{R}^+$ is multiplicative

Recipe:

1. transform your data: $\log(x)$
2. analyse your data
3. transform desired parameters back: e^x

Frequency Distribution of suspension load of a river



Recipe 2: lower limit < x < upper limit

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- **Step 1: Check the distribution** of the data using histograms, boxplots, or Q-Q plots to assess skewness and identify potential outliers. If the data are proportions, ensure that all values are between 0 and 1. If there are values of 0 or 1, consider applying a small adjustment (e.g., adding a small constant to 0 values and subtracting a small constant from 1 values) to avoid issues with the logit transformation.
- **Step 2a: (optional) If your data has a lower and/or upper limit** that is **not 0 or 1**, perform a min-max normalization to scale the data to the]0,1[interval before applying the logit transformation. Therefore you have to apply the following transformation to your data:

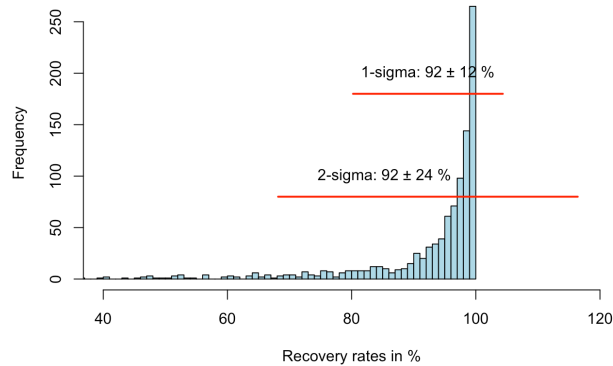
$$x' = (x - \min(x)) / (\max(x) - \min(x)).$$

- **Step 2: Apply the logit transformation** to the data using the formula $x' = \log(x/(1-x))$, where x is the original data and x' is the transformed data. Hereby, exclude the minimum and maximum values to avoid issues like zero and zero divisor.
- **Step 3:** After logit transformation, **check the distribution** of the transformed data to ensure that it is more symmetric and closer to a normal distribution. If there is still a considerable skewness, you may consider applying e.g. a square-root, an arctan() or an arcsin() transformation data. If the data is symmetric but with an obvious smaller or larger variance, you may consider applying a scaling.
- **Step 4: Calculate the arithmetic mean and standard deviation on the logistic scale**, which are now appropriate measures. Do not interpret these results. Just calculate the 1-sigma-ranges on the logistic scale as mean $\pm$ standard deviation in the logistic FS.
- **Step 5: Back-transform the results to the original scale** by applying the inverse logit function to the mean and both bounds of the 1-sigma interval of the logit-transformed data.
- Due to the nature of the logit transformation, these back-transformed values will always be between 0 and 1, ensuring that they are meaningful for proportions and compositional data.
- **Step 5b: (optional) If you performed a min-max normalization in Step 2a**, back-transform the results to the original scale by applying the inverse of the min-max normalization.

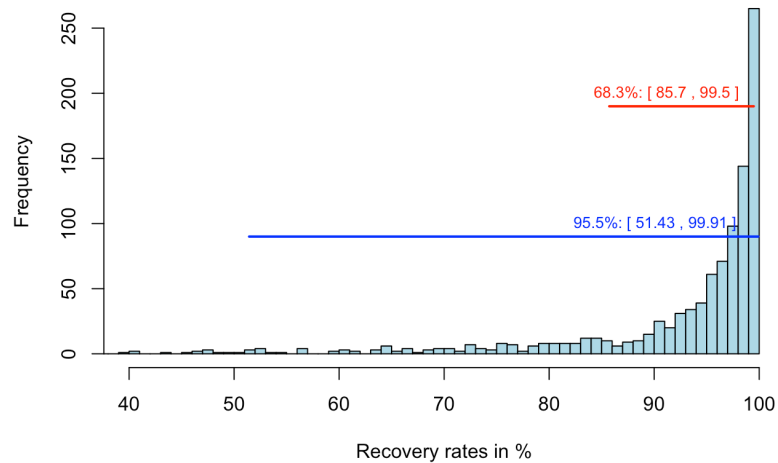
Recipe 2: lower limit < x < upper limit

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Percent area recovered within 30 years

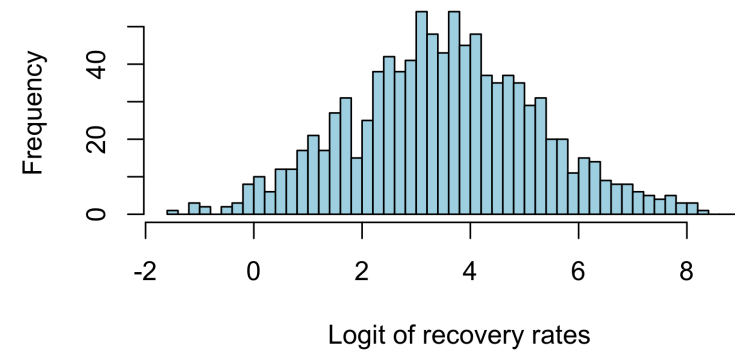


Percent area recovered within 30 years



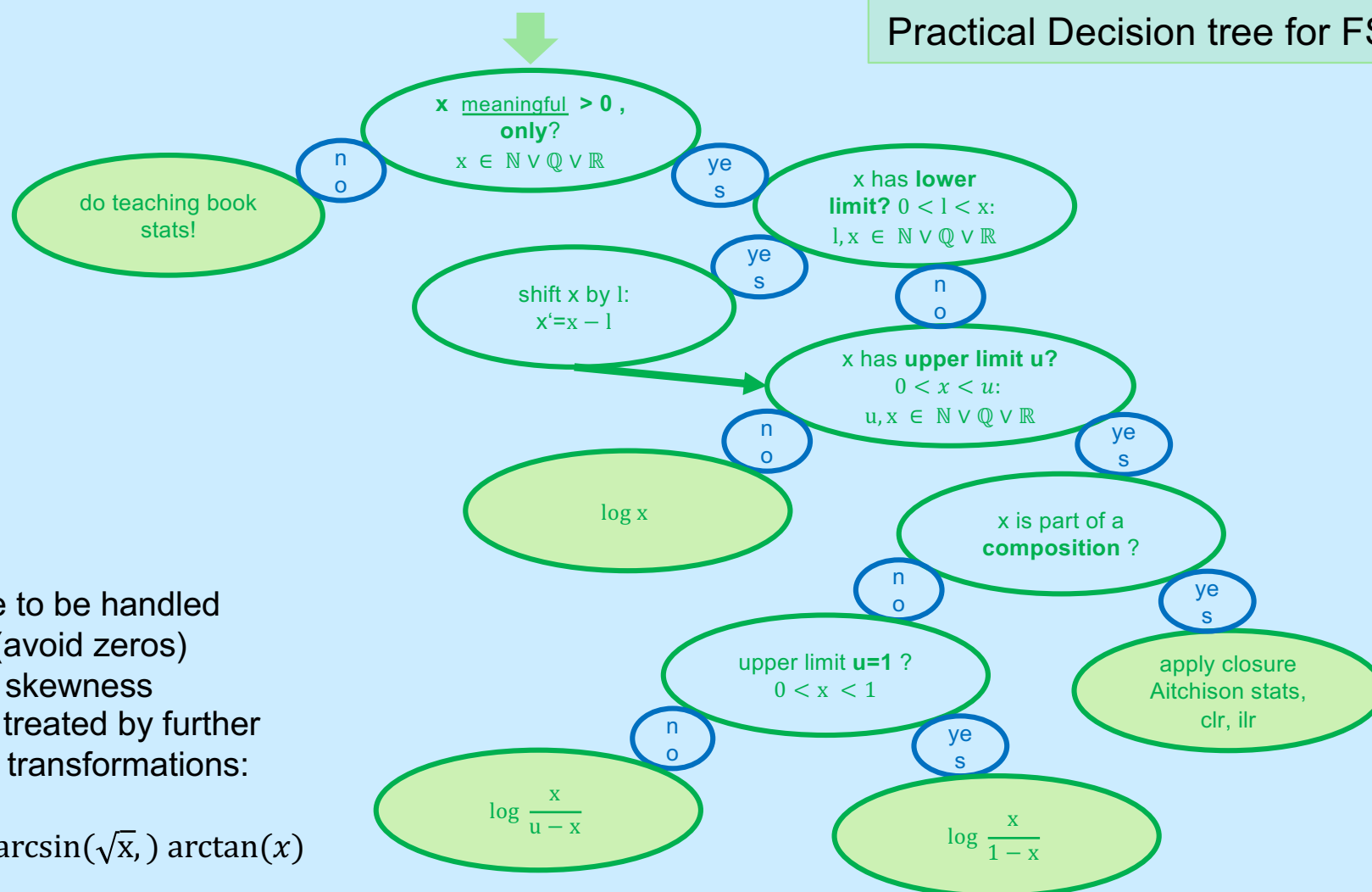
$$y' = \log\left(\frac{y}{1-y}\right)$$

Logit-transformed data



$$y = \frac{e^{y'}}{1 + e^{y'}}$$

Practical Decision tree for FST



Remarks:

- limits have to be handled with care (avoid zeros)
- remaining skewness should be treated by further non-linear transformations:
e.g.
 $\arcsin(x)$, $\arcsin(\sqrt{x})$, $\arctan(x)$

Other useful transformations:

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!! For shape control only: They are not capable of solving algebraic constraints !!

1. Square root: $y' = \sqrt{y}$ positive skewness and kurtosis, variable remains in $\mathbb{R}^+$
2. Reciprocal: $y' = \frac{1}{y}$ positive skewness and kurtosis, variable remains in either in $\mathbb{R}^+$ or $\mathbb{R}^-$
3. Arcsine & Arcsine-square root: $y' = \arcsin y$ & $y' = \frac{2}{\pi} \arcsin \sqrt{y}$ positive skewness in $]0,1[_{\mathbb{R}^+}$
4. Box-Cox-Transformation for $\frac{x^\lambda - 1}{\lambda}, \lambda \neq 0 \wedge x > 0$

Only in combination with log and logistic-transformation:

5. upper boundary: $y' = \max - y$. negative skewness in combination with e.g. log; from $\mathbb{R}^+]0, u[$ to $\mathbb{R}^+]0,1[$
6. Adding constant: $y' = y + a$. translation for e.g. avoiding 0-value for log or reciprocal

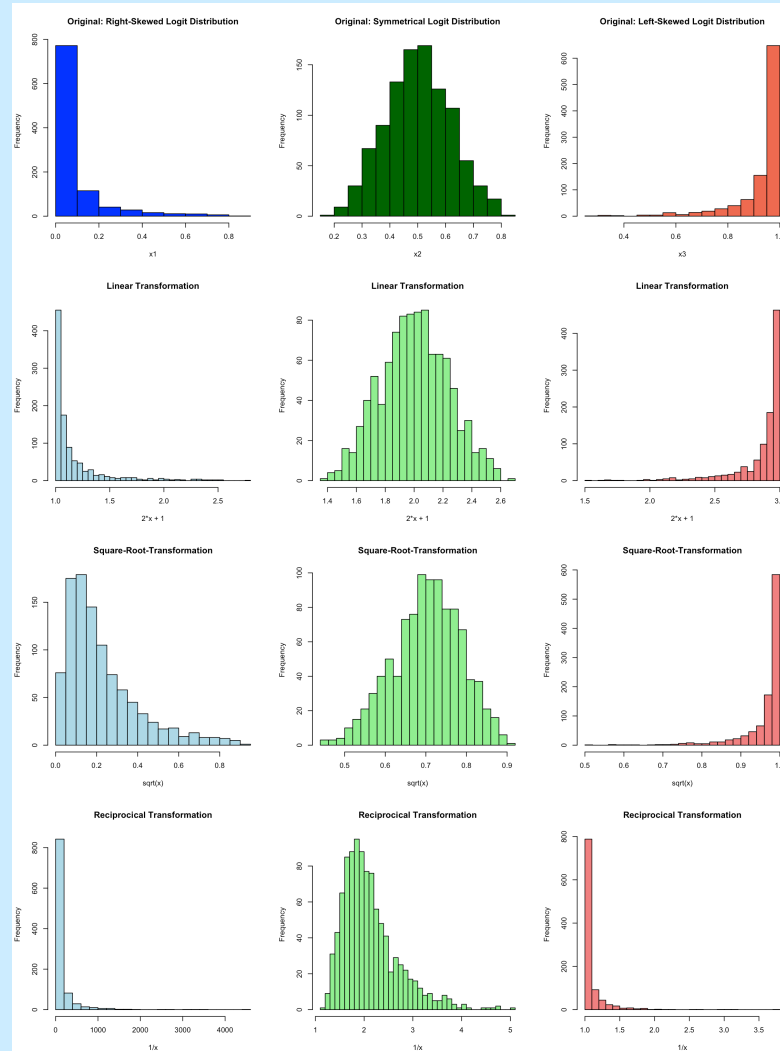
plus **standardization** and **normalization** technics but do not help for non-normality, skewness or algebraic constraints!!! (an overview is given e.g. [Minkyungs'sblog](#))

cf. also [Zhang et al, 2024](#)

Additional transformations

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original



$$x^2 + 1$$

$$\sqrt{x}$$

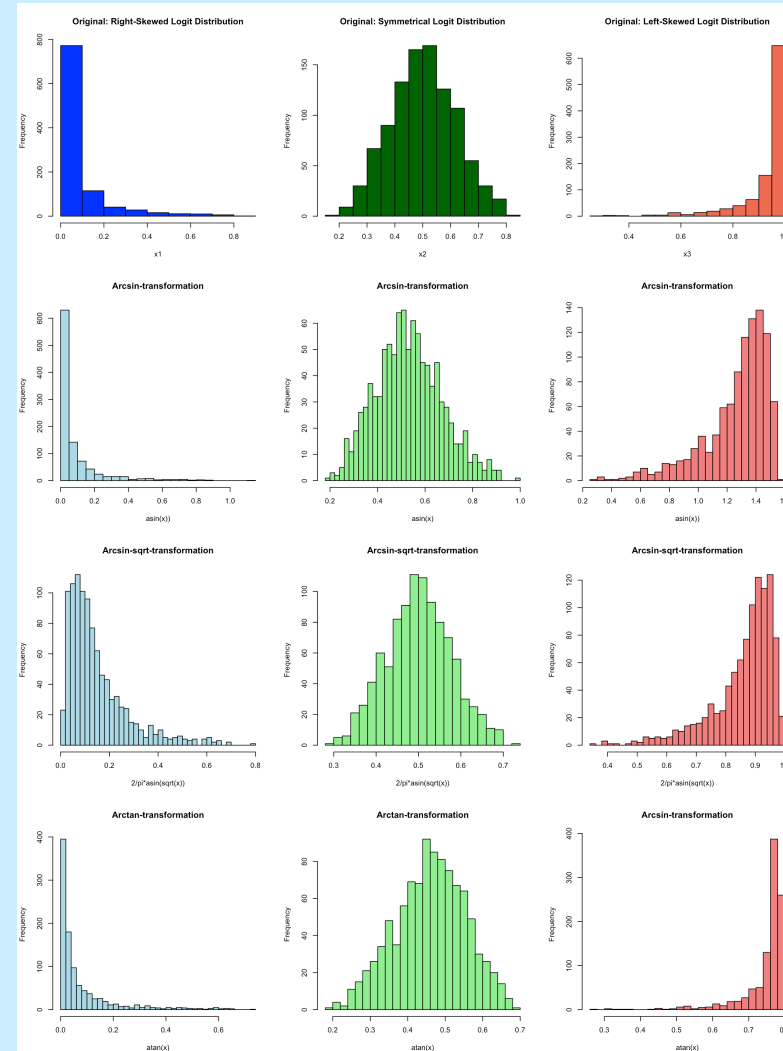
$$1/x$$

original

asin x

$$\frac{2}{\pi} \text{asin} \sqrt{x}$$

atan x



4. Recipes: Preparing real-world data for ML

Additional transformations

60

original

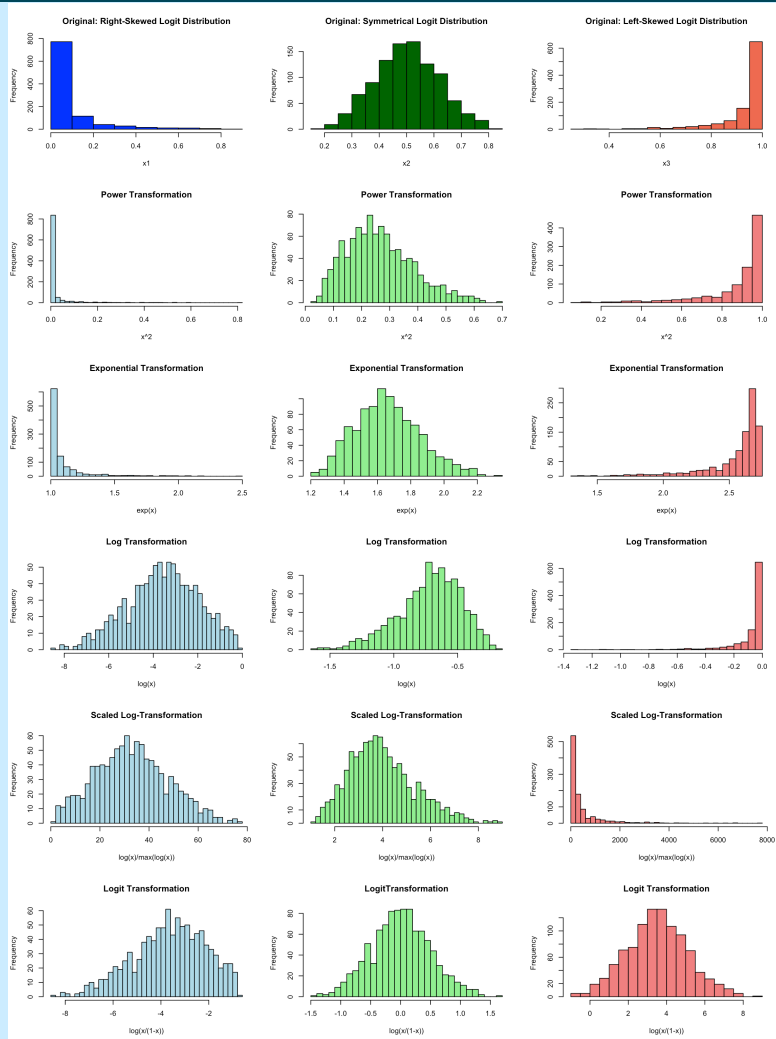
x^2

e^x

$\log x$

$\frac{\log x}{\max(\log x)}$

$\frac{x}{1-x}$



Here, only the **logit-transformation** preserves and produces **symmetry**!

4. Recipes: Preparing real-world data for ML

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Hands-on exercise

Open the exercise file “exercise_EGU26_SC2-28”
as **R-Markdown** or **Jupyter-Notebook**
at our Git-hub repo

https://github.com/soga-lab/EGU2026_SC

Work and discuss in small groups of 4-5 people!



Thanks for your attention!

For further feedback or questions:
soga@zedat.fu-berlin.de